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Executive Summary

The portfolio consultation evaluates a diversified portfolio consisting of three Philippine equities: PLDT Inc. (TEL), Manila Electric Company (MER), and Aboitiz Equity Ventures (AEV). In addition, it includes international technology equities NVIDIA Corporation (NVDA) and Meta Platforms, Inc. (META) as well as Bitcoin (BTC). Using daily market data from January 2020 through December 2025, the study assesses portfolio performance, diversification, optimization, and downside risk. Furthermore, the SPDR S&P 500 ETF Trust (SPY) serves as the benchmark while the Philippine 91-day Treasury Bill serves as the risk-free rate.

The empirical results demonstrate that the portfolio substantially outperformed the benchmark. However, this out-performance came with significantly greater risk. The equal-weighted Buy-and-Hold strategy generated a 743.04% cumulative return and a 42.83% CAGR. In comparison, SPY generated a 129.06% cumulative return and a 14.87% CAGR. Nevertheless, the Buy-and-Hold strategy recorded 35.12% annualized volatility and a -58.21% maximum drawdown. Meanwhile, monthly rebalancing reduced annualized volatility to 24.18% and maximum drawdown to -43.20%. However, cumulative returns also declined to 384.85%.

Moreover, the downside-risk analysis highlights the trade-off between return maximization and capital preservation. During major market stress events, the GMV portfolio consistently experienced smaller losses than the Maximum Sharpe portfolio. For instance, during the 2020 COVID-19 crash, GMV declined by 22.55% compared with 28.83% for Maximum Sharpe. Similarly under the hypothetical 1.5x Black Swan scenario, losses were 35.89% and 45.30%, respectively.

Therefore, portfolio selection should reflect the investor's risk tolerance and objectives. The Maximum Sharpe portfolio suits investors seeking higher returns despite greater risk. Conversely, the GMV portfolio suits risk-averse investors prioritizing capital preservation. For a balance between growth and risk control, monthly rebalancing provides a disciplined approach. It reduces volatility and drawdown while still substantially outperforming the benchmark.

1 Introduction

Portfolio management involves allocating funds among different categories of assets to ensure the proper achievement of a balance between expected return and risk. Modern portfolio theory provides the base theoretical foundation for this process, arguing that investors should classify and evaluate securities portfolios not solely on their individuality, but also on expected returns and volatility, as well as their covariance with other assets in the portfolio, as argued by Markowitz (1952). Thus, diversification may assist investors in reducing portfolio risk without an equivalent reduction in expected return when selected assets are imperfectly correlated.

This case study sheds light on evaluating a diversified portfolio consisting of three Philippine equities: PLDT Inc. (TEL), Manila Electric Company (MER), and Aboitiz Equity Ventures (AEV); two international technology equities: NVIDIA Corporation (NVDA) and Meta Platforms, Inc. (META); and one alternative digital asset: Bitcoin (BTC). Based on these assets, the portfolio combines domestic Philippine securities, Western growth-oriented technology stocks, and a cryptocurrency. This allocation of diverse asset classes provides a structured opportunity to assess whether diversification across different platforms, industries, asset classes, and geographies enhances portfolio performance. However, each individual asset must be assessed carefully as certain assets exhibit substantial volatility and diversification benefits vary across market regimes and periods of financial stress (Almeida & Gonçalves, 2023).

Through the use of daily adjusted closing price data from January 2, 2020 through December 31, 2025, this study gauges the return and risk profile of the investment portfolio benchmarked against the SPDR S&P 500 ETF Trust (SPY), which reflects the performance of large-capitalization equities from the United States. Furthermore, daily log return measurements are calculated to examine descriptive statistics, return distributions, correlations, and volatility among the selected assets. The empirical analysis compares portfolio execution with and without monthly rebalancing to assess whether restoring portfolio weights periodically affects portfolio growth, risk exposure, and overall performance.

Moreover, the portfolio is evaluated using the Philippine 91-Day Treasury Bill as the proxy for the risk-free rate. This rate supports the computation of the Sharpe ratio, measuring excess return earned per unit of total risk (Sharpe, 1966). In addition, mean-variance optimization is applied to construct the efficient frontier and identify both the Global Minimum Variance portfolio and the Maximum Sharpe Ratio portfolio. These optimization outputs are compared with alternative asset allocations to determine which strategy offers the most favorable risk-return trade-off across the sample period.

Additionally, because standard deviation and average returns cannot fully capture extreme losses, this study conducts a comprehensive downside risk assessment. Non-parametric Historical Value at Risk, Expected Shortfall (Conditional Value at Risk), Parametric Gaussian Value at Risk, Monte Carlo simulations, and crisis stress testing are utilized to evaluate potential portfolio losses under normal and adverse market conditions. Expected Shortfall is particularly essential because it measures the conditional average loss beyond the Value at Risk threshold, providing deeper insight into severe tail-risk exposure (Acerbi & Tasche, 2002).

Ultimately, the study aims to develop an evidence-based investment recommendation for an institutional investor considering exposure to Philippine equities, Bitcoin, and international technology equities. The recommendations synthesize empirical return distributions, correlation structures, optimized frontier allocations, benchmark-relative performance, and portfolio resilience across historical stress regimes, establishing a defensible balance between return potential, total volatility, and downside risk tolerance.

1.1 Mandate and Assigned Asset Universe

This report delivers a quantitative portfolio management consultation for Group 3 (Charlie). The assigned asset universe comprises six multi-asset constituents across four distinct asset classes:

- 1. Philippine Equities (Domestic Core):** PLDT Inc. (TEL), Manila Electric Company (MER), and Aboitiz Equity Ventures (AEV).

2. **International Equities (Growth Satellite):** NVIDIA Corporation (NVDA) and Meta Platforms, Inc. (META).
3. **Digital Assets (Alternative Satellite):** Bitcoin (BTC).
4. **Benchmark ETF:** SPDR S&P 500 ETF Trust (SPY).
5. **Risk-Free Rate Baseline:** Philippine 91-Day Treasury Bill secondary market rate ($R_f = 5.25\%$ per annum benchmark).

1.2 Risk-Free Rate Benchmark Specification

The hurdle rate for all excess return, Sharpe ratio, and tangency portfolio calculations is anchored to the Philippine 91-Day Treasury Bill (3-Month T-Bill) secondary market yield. Quoted at an annualized benchmark rate of $R_f = 5.25\%$, it is converted into a daily log-equivalent rate assuming a 252-trading-day annual convention:

$$r_{f,\text{daily}} = \frac{\ln(1 + R_f)}{252} = \frac{\ln(1 + 0.0525)}{252} \approx 0.00020307 \quad (0.0203\% \text{ per day}) \quad (1)$$

This hurdle rate represents the riskless baseline return earned by domestic institutional capital.

1.3 Report Organization

The remainder of this report is structured as follows: Section 2 details data acquisition, FX conversion, calendar alignment, and missing value treatment. Section 3 conducts exploratory financial analysis and distributional diagnostics. Section 4 evaluates Buy-and-Hold versus Monthly Rebalanced equal-weighted portfolios. Section 5 solves for the Efficient Frontier, Global Minimum Variance (GMV), and Maximum Sharpe Ratio portfolios. Section 6 assesses downside risk via VaR, Expected Shortfall, Monte Carlo simulations, and stress testing. Section 7 presents executive investment recommendations, and Section 8 lists references.

2 Data Acquisition and Preparation

The construction of a quantitative multi asset portfolio requires empirical rigor during data collection, market calendar alignment, missing value interpolation, and feature engineering. This section details the complete end-to-end data pipeline executed for Group 3, tracing raw multi-source ingestion through final matrix transformations.

2.1 Asset Universe and Ingestion Pipeline

The portfolio universe spans seven financial assets across four distinct asset classes, alongside the benchmark index and domestic risk free rate. The sampling horizon spans January 2, 2020 through December 31, 2025 (the six calendar years 2020–2025), yielding a master panel of 1,508 daily trading dates per asset.

Raw asset prices were programmatically ingested via automated Python REST scripts leveraging the `yfinance` API (Yahoo Finance, 2026) for US-listed and cryptocurrency instruments. Philippine equities are sourced from the official PSE EDGE portal (Philippine Stock Exchange, 2026) (MER, AEV), with PLDT represented via its NYSE-listed ADR, then converted to US dollars to preserve a single currency denomination across the portfolio.

- **PLDT Inc. (TEL):** Represented via its NYSE-traded ADR (PHI) as an accessible market-data proxy for TEL, with the US trading calendar and USD denomination as implications.
- **Manila Electric Company (MER):** Primary PSE listing, daily OHLCV from PSE EDGE, converted from PHP to USD.
- **Aboitiz Equity Ventures (AEV):** Primary PSE listing, daily OHLCV from PSE EDGE, converted from PHP to USD.
- **NVIDIA Corporation (NVDA):** Ingested directly from NASDAQ exchange endpoints.
- **Meta Platforms, Inc. (META):** Ingested directly from NASDAQ exchange endpoints.

- **Bitcoin (BTC):** Ingested from global cryptocurrency spot market feeds as BTC-USD.
- **SPDR S&P 500 ETF Trust (SPY):** Ingested from NYSE Arca as the benchmark ETF.
- **Risk-Free Rate:** Annualized Philippine 91-Day Treasury Bill secondary market rate (5.25 percent benchmark) (Bangko Sentral ng Pilipinas, 2026).

Table 1 presents the complete data acquisition specifications, exchange mappings, observation counts, and sample horizons.

Table 1: Asset Universe and Data Acquisition Specifications

Asset Ticker	Asset Class	Exchange / Source	Observations	Sample Period
TEL	Domestic Equity	PSE / NYSE (PHI)	1508 Days	Jan 2, 2020 – Dec 31, 2025
MER	Domestic Equity	PSE / EDGE	1422 Days	Jan 2, 2020 – Dec 31, 2025
AEV	Domestic Equity	PSE / EDGE	1421 Days	Jan 2, 2020 – Dec 31, 2025
NVDA	International Equity	NASDAQ	1508 Days	Jan 2, 2020 – Dec 31, 2025
META	International Equity	NASDAQ	1508 Days	Jan 2, 2020 – Dec 31, 2025
BTC	Cryptocurrency	Global Crypto	1508 Days	Jan 2, 2020 – Dec 31, 2025
SPY	Benchmark ETF	NYSE Arca	1508 Days	Jan 2, 2020 – Dec 31, 2025
PH 91D T-Bill	Risk-Free Rate	BSP / BTr	1508 Days	Jan 2, 2020 – Dec 31, 2025

2.2 Calendar Alignment, Relational Joins, and Imputation

Financial asset price series originate from heterogeneous trading environments operating on asynchronous market schedules. Cryptocurrency markets trade continuously 24 hours a day across 365 days a year, whereas US equity exchanges observe NYSE holiday schedules and Philippine equities trade on PSE operating calendars.

To reconcile calendar discrepancies, individual asset time series were joined using relational outer joins keyed on ISO-8601 standardized date timestamps. The master date index was anchored to the official 1,508 trading day calendar of the SPDR S&P 500 ETF Trust (SPY).

Philippine closing prices from PSE EDGE are quoted in Philippine pesos and are converted to US dollars using the daily PHP per USD spot rate (WM/Refinitiv via the USDPHP=X feed). The rate observed on each PSE trading date is applied to that date’s prices, with the most recent published rate carried forward when the FX feed does not publish on a PSE-only trading day. This keeps the entire portfolio, including the SPY benchmark, in a single currency.

Missing observations are handled explicitly by instrument type. For the US-listed TEL ADR, a zero-volume row from the vendor carries no new price signal and is stored as missing rather than carried forward; treating a stale quote as a settlement price would manufacture zero-return observations and distort variance, correlation, and tail-risk estimates. For the PSE-listed instruments, dates absent from the PSE trading calendar are naturally missing in the master panel. Beyond Philippine holidays falling on US trading days, these absences reflect genuine market closures over the sample: the two-day suspension of March 17–18, 2020; calamity-driven closures (Taal volcano ashfall on January 13, 2020, Typhoon Ulysses on November 12, 2020, Super Typhoon Karding on September 26, 2022, the suspension of July 24, 2024, and election-day closure on May 12, 2025); and a trading cancellation from a systems glitch on January 4, 2022. One further date, October 24, 2022, is missing for both names even though the PSE was open that day, reflecting a gap in the captured PSE EDGE records rather than a market closure; October 25, 2022 is missing for AEV alone, a per-stock no-trade day on which MER printed a single flat trade. No imputation is applied to the stored panel: missing rows remain missing, and the ingestion script reports per-asset traded-day counts and missing-day totals on every refresh. At the analysis layer, wealth aggregation and backtesting carry the last observed price across no-trade days so that the resumption-day return captures the full gap move, while descriptive statistics and correlations are computed on observed returns only; no-trade days are never assigned a zero return.

2.3 Feature Engineering and Mathematical Transformations

Raw adjusted closing prices were subjected to mathematical transformations to construct stationary return series, scaled benchmark features, cumulative growth paths, and risk parameters required for downstream quantitative modeling.

2.3.1 Logarithmic Return Transformation

Adjusted closing prices were transformed into continuously compounded daily log returns according to the standard logarithmic return formula.

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) = \ln(P_{i,t}) - \ln(P_{i,t-1}) \quad (2)$$

Logarithmic transformation converts compounding price ratios into linear additive values over time, facilitating multi period horizon aggregation. Log transformation also stabilizes return variance across high growth regimes, ensuring that extreme price movements in assets like Bitcoin or NVIDIA remain scale comparable with lower-volatility equities.

2.3.2 Risk-Free Rate Daily Scaling

The domestic benchmark risk free rate is quoted as an annualized yield percentage of 5.25 percent per annum. To align the risk free rate with daily log return series, the annual yield was transformed into a daily log equivalent rate using a 252 trading day annual convention.

$$r_{f,t} = \frac{\ln(1 + R_f^{\text{annual}})}{252} = \frac{\ln(1 + 0.0525)}{252} \approx 0.00020307 \quad (3)$$

This transformation yields a constant daily risk free baseline of 0.0203 percent per day, which serves as the hurdle rate in subsequent Sharpe ratio and excess return computations.

2.3.3 Cumulative Wealth Index Feature

To track total capital appreciation from an initial baseline of USD 1.00, a cumulative wealth index feature was constructed for each asset by exponentiating cumulative log returns.

$$W_{i,t} = \exp \left(\sum_{\tau=1}^t r_{i,\tau} \right) = \prod_{\tau=1}^t \left(\frac{P_{i,\tau}}{P_{i,\tau-1}} \right) \quad (4)$$

The cumulative wealth feature provides an intuitive trajectory of asset growth without distortion from daily volatility oscillations.

3 Exploratory Financial Analysis

Exploratory financial analysis establishes the empirical foundation required to construct robust portfolio optimization algorithms and risk management frameworks. This section evaluates central tendency, volatility dynamics, distributional shape, tail risk diagnostics, and pairwise asset return co-movements across the 1,507 daily log return rows of the master panel spanning January 2, 2020 through December 31, 2025. Assets with a complete price record (TEL, NVDA, META, BTC, SPY) contribute 1,507 observed daily returns, while the PSE-sourced instruments contribute 1,421 (MER) and 1,420 (AEV) observed returns, the shortfall reflecting days without captured PSE prices: Philippine holidays falling on US trading days, genuine market closures (the two-day suspension of March 17–18, 2020, calamity- and election-driven closures, and a systems-glitch cancellation on January 4, 2022), a single per-stock no-trade day for AEV on October 25, 2022, and one source-data gap on October 24, 2022, when the PSE was open but no PSE EDGE records were captured for either domestic name. All statistics below are computed on observed returns only; no-trade days are never assigned a zero return, and returns that span a no-trade gap are measured against the last traded close so the full move appears in the sample.

3.1 Empirical Descriptive Statistics and Risk Return Profiles

Table 2 summarizes the descriptive statistics computed from the daily log return series across all seven assets.

Table 2: Exploratory Financial Descriptive Statistics (Jan 2, 2020 – Dec 31, 2025)

Metric	TEL	MER	AEV	NVDA	META	BTC
Observations	1507.0000	1421.0000	1420.0000	1507.0000	1507.0000	1507.0000
Minimum	-0.2703	-0.1668	-0.1680	-0.2040	-0.3064	-0.4647
Mean	0.0003	0.0003	-0.0005	0.0023	0.0008	0.0017
Maximum	0.1548	0.1175	0.1258	0.2181	0.2093	0.1915
Stdev	0.0199	0.0204	0.0253	0.0333	0.0279	0.0393
Skewness	-1.7677	-0.2346	-0.2215	0.0439	-1.0740	-1.2216
Kurtosis	31.4634	5.7962	3.7706	4.0527	22.0895	16.2926

The results reveal pronounced divergence in expected return and risk parameters across asset classes. NVIDIA Corporation (NVDA) generated the highest average daily log return at 0.0023, an annualized mean of roughly 57.5 percent. Bitcoin (BTC) followed at 0.0017 per day (42.3 percent annualized), with Meta Platforms (META) and the SPDR S&P 500 ETF Trust (SPY) recording moderate means of 0.0008 and 0.0005. The Philippine universe displayed subdued central tendencies: Manila Electric Company (MER) averaged 0.0003 per day (8.0 percent annualized) and PLDT Inc. (TEL) 0.0003 (8.0 percent). Aboitiz Equity Ventures (AEV) was the only asset with a negative mean return, at -0.0005 per day (-13.6 percent annualized).

Daily volatility metrics confirm the expected risk hierarchy. Bitcoin exhibited the highest daily standard deviation of 0.0393 (62.3 percent annualized), followed by NVIDIA at 0.0333 (52.9 percent) and Meta at 0.0279 (44.2 percent). Aboitiz Equity Ventures registered the most volatile Philippine series at 0.0253 (40.2 percent annualized), while the lower-volatility domestic names TEL (PSE Services) and MER (PSE Industrial) recorded 0.0199 (31.5 percent annualized) and 0.0204 (32.3 percent annualized) respectively. The benchmark SPY displayed the lowest daily volatility at 0.0131 (20.8 percent annualized), reflecting the risk reduction achieved through broad market diversification.¹

Third and fourth statistical moments document departure from Gaussian normality, with severity that is heterogeneous across the universe. Skewness is negative for TEL (-1.7677), BTC (-1.2216), META (-1.0740), SPY (-0.5584), and mildly negative for the PSE names once gap-crossing moves are included, MER (-0.2346) and AEV (-0.2215); NVDA is approximately symmetric (0.0439). Negative skew indicates that extreme negative drawdowns occur with greater magnitude than positive surges for those assets.

Kurtosis is reported throughout in its excess form, with a Gaussian baseline of zero. All seven assets are leptokurtic, but the domestic instruments are far closer to the Gaussian benchmark than the global block. TEL is the most extreme at 31.4634, driven largely by idiosyn-

¹PSE EDGE closing prices are unadjusted for corporate actions. Ex-dividend days therefore appear as small single-day price dips, on the order of 0.5 percent per quarterly dividend for MER, which marginally inflates measured daily volatility and kurtosis for the domestic names.

cratic single-day events, notably the -27.0 percent daily return of December 19, 2022. META (22.0895), BTC (16.2926), and SPY (13.3891) follow, while MER (5.7962), NVDA (4.0527), and AEV (3.7706) exhibit moderate excess kurtosis. The heavy tails confirm that standard mean-variance assumptions understate tail risk, but the corrected PSE-based series show that the domestic names are materially better behaved than the earlier vendor-sourced data suggested.

3.2 Historical Return Trajectories and Volatility Clustering

Figure 1 illustrates the historical daily log return series for each asset across the sampling period.

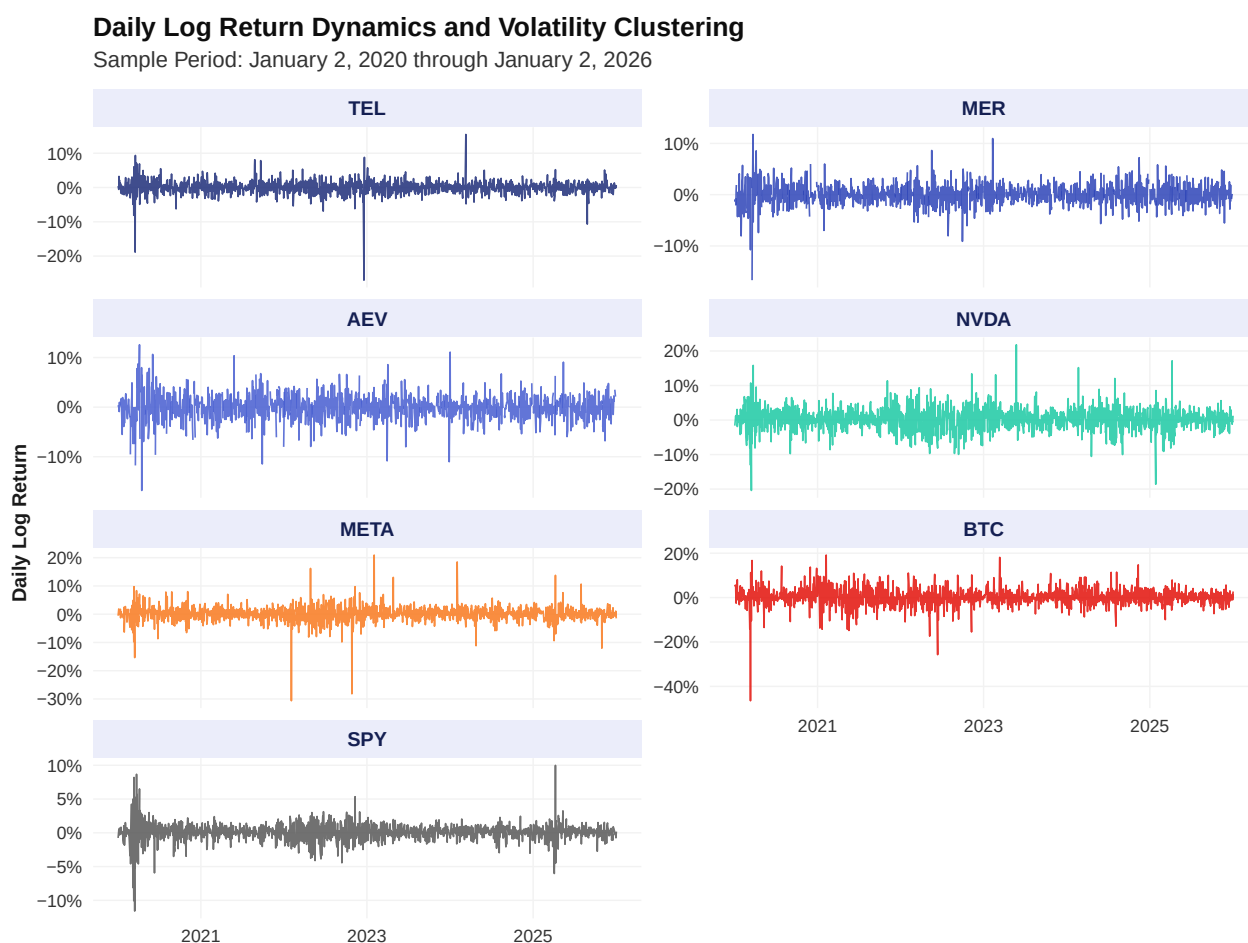


Figure 1: Daily Log Return Trajectories Across Portfolio Assets (Jan 2020 – Dec 2025)

The trajectories exhibit pronounced volatility clustering: large return shocks of either sign are followed by large shocks, while tranquil periods persist in distinct bands. The March 2020

systemic liquidity shock was the defining event of the sample. Measured close-to-close from February 19 to March 23, 2020, the crisis losses were severe and uneven: TEL -9.06 percent, MER -19.37 percent, AEV -40.82 percent, NVDA -28.24 percent, META -32.00 percent, BTC -36.74 percent, and SPY -33.40 percent. The PSE-sourced series show honest gaps rather than fabricated flat days on dates without captured prices: Philippine holidays falling on US trading days, the two-day PSE market suspension of March 17–18, 2020, AEV’s sole per-stock no-trade day on October 25, 2022, and a source-data gap on October 24, 2022. The exchange reopened on March 19 with shortened trading hours, and both MER and AEV printed prices that day; their re-opening log returns, measured against the last pre-suspension close, capture the full gap moves (MER -16.68 percent, AEV -11.75 percent). The largest domestic single-day extremes of the sample were AEV’s -16.8 percent trough on April 16, 2020, MER’s -16.7 percent re-open on March 19, 2020, and TEL’s -27.0 percent on December 19, 2022.

Sector-specific regimes followed in subsequent years. The 2022 monetary tightening cycle produced sustained dispersion in technology and crypto assets, with Meta posting a -30.6 percent single-day return on February 3, 2022 after its earnings release and calendar-year losses of -64.2 percent for both META and BTC and -50.3 percent for NVDA. The domestic names absorbed the same cycle far more gently, with calendar-year 2022 returns of -31.7 percent for TEL, -7.6 percent for MER, and -3.3 percent for AEV. TEL’s idiosyncratic -27.0 percent shock of December 19, 2022 stands out as the largest domestic move of the sample and is visible in the lower tail of its return distribution.

3.3 Distributional Shape and Non-Normality Diagnostics

Figure 2 compares the empirical kernel density distributions of daily log returns against fitted Gaussian reference curves.

Empirical Distribution Shape: Heavy Tails and Non-Normality

Comparison of empirical kernel density against theoretical normal distribution

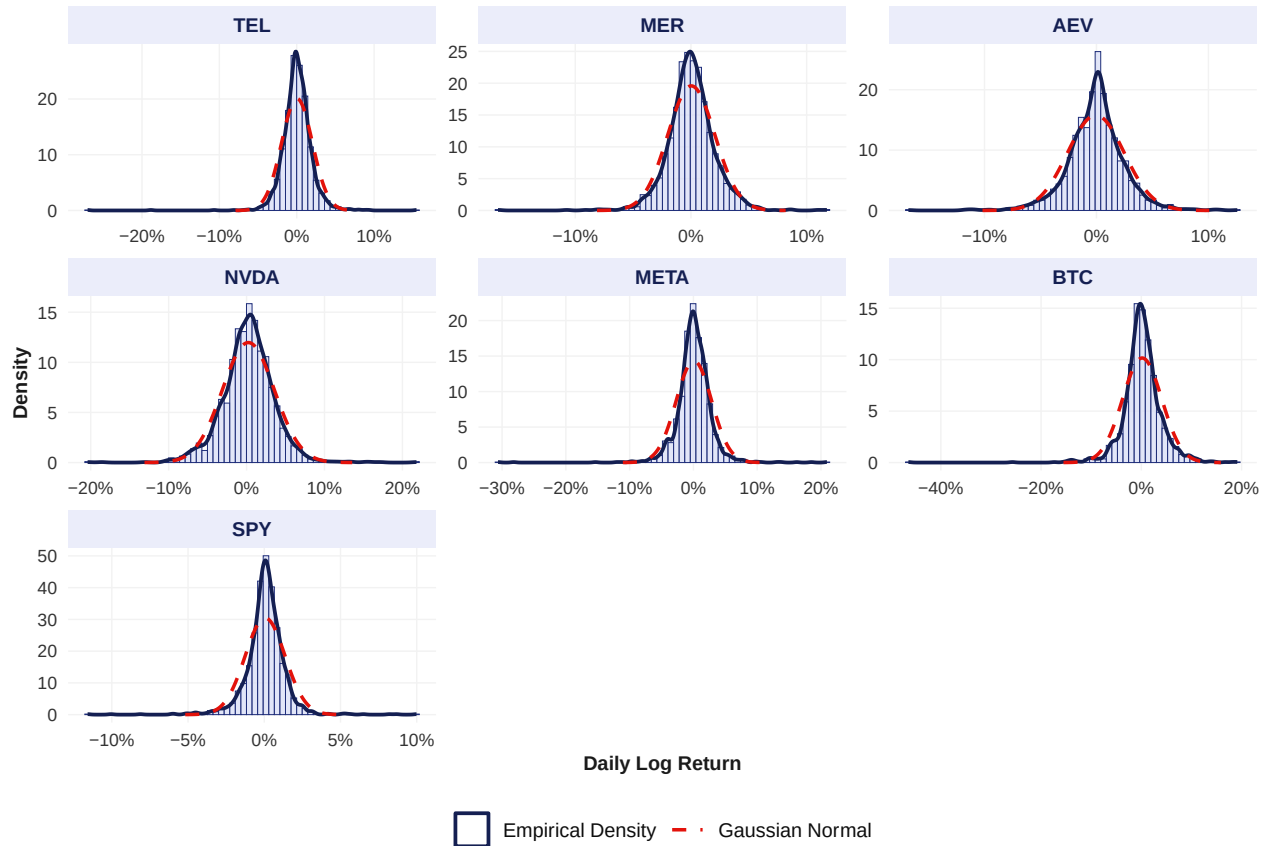


Figure 2: Empirical Return Distributions vs. Gaussian Reference Curves

The empirical density overlays show the classic departure pattern: central peaks that rise above the theoretical normal, shoulders that collapse faster than the Gaussian, and tails that extend much farther outward. The effect is strongest for TEL, BTC, META, and SPY, consistent with their excess kurtosis values, and mildest for NVDA, AEV, and MER, whose empirical densities track the Gaussian far more closely while still exhibiting wider tails than the reference curve.

Figure 3 provides normal Quantile-Quantile (Q-Q) diagnostics for tail behavior.

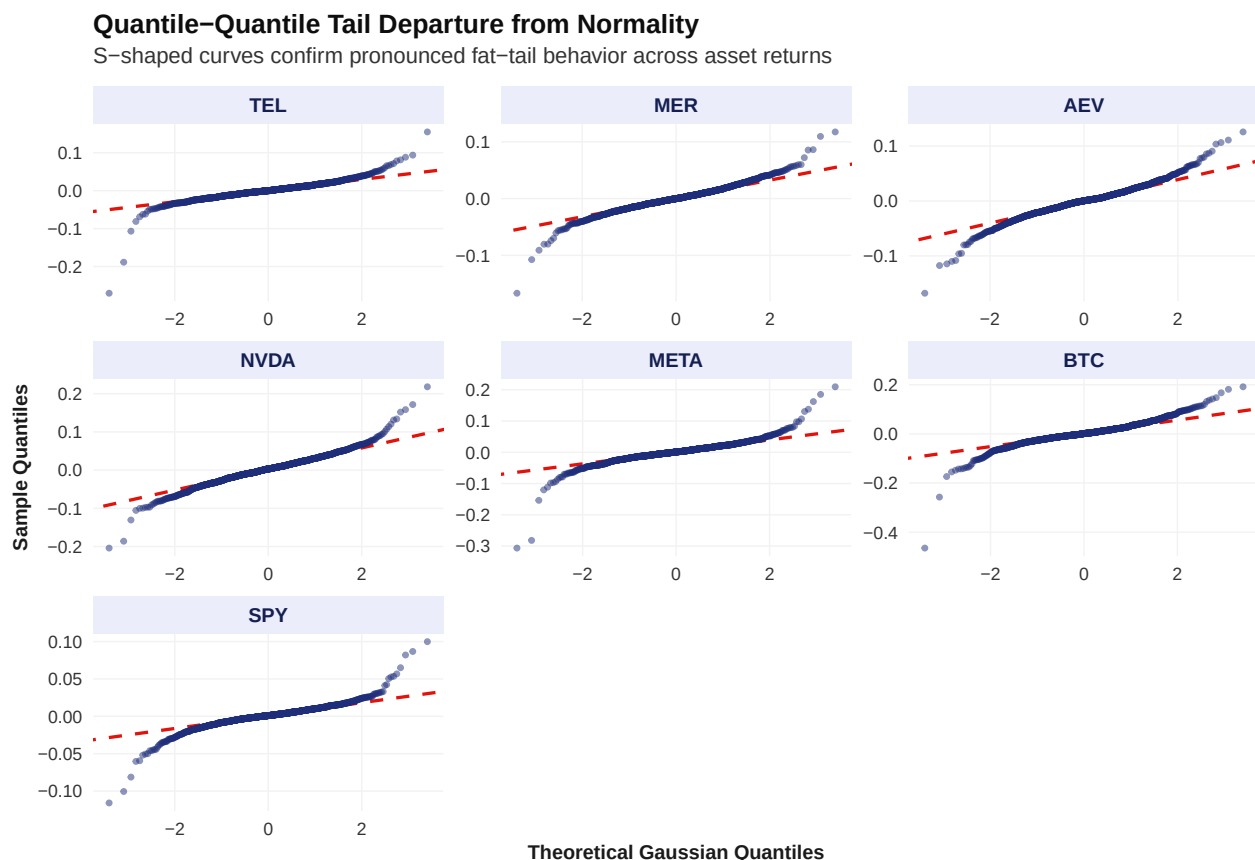


Figure 3: Normal Q-Q Diagnostics Demonstrating Heavy Tail Leptokurtosis

The Q-Q plots display S-shaped departures from the theoretical reference line across the universe. Sample quantiles bend sharply below the line in the lower left quadrant and above it in the upper right, confirming that realized tails are fatter than normality implies. The curvature is most pronounced for TEL and the global block, while NVDA, AEV, and MER show milder, though still visible, tail divergence. Risk models that assume Gaussian returns will underestimate tail risk across the portfolio, and the effect is largest in the technology and cryptocurrency positions.

3.4 Return Dispersion and Extreme Outlier Comparison

Figure 4 presents cross-asset return dispersion boxplots, illustrating medians, interquartile ranges (IQR), and extreme tail outliers.

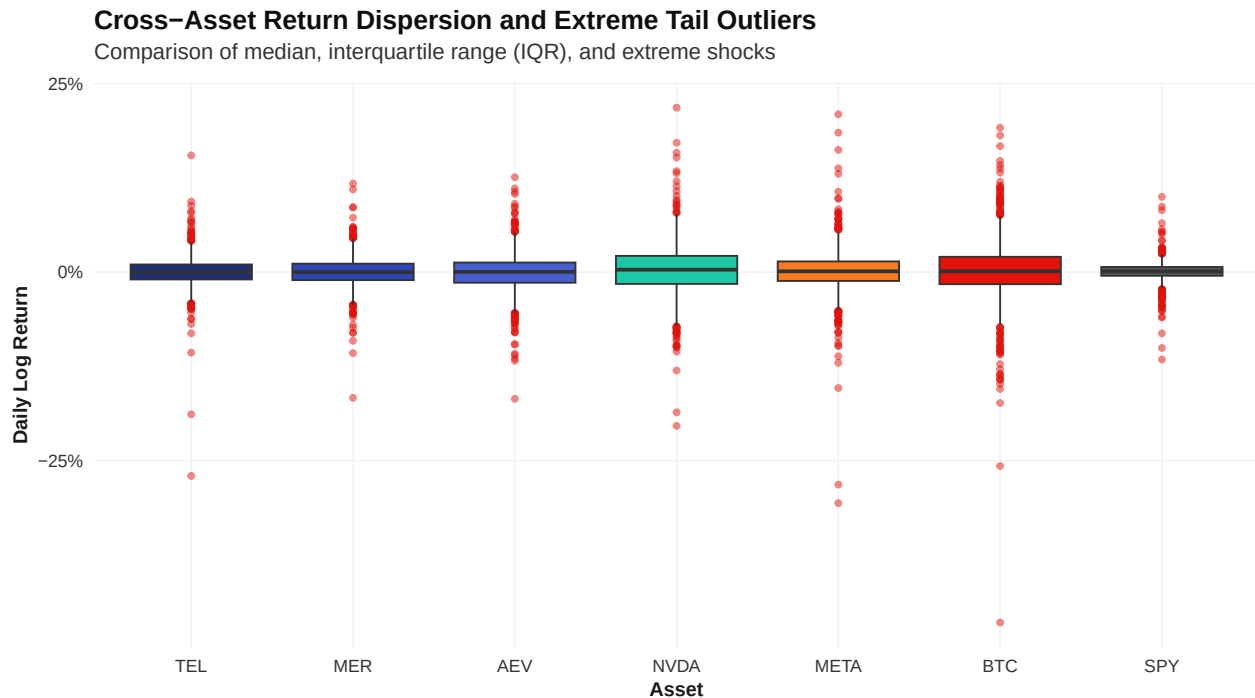


Figure 4: Cross-Asset Return Dispersion and Extreme Outlier Boxplots

The boxplots highlight distinct groupings in return dispersion. SPY displays the most compact interquartile box and the shortest whiskers, reflecting index-level diversification. The Philippine names exhibit narrow IQRs bounded approximately within ± 1 percent daily, with outliers that are modest relative to the global block: TEL spans -27.0 percent to +15.5 percent, AEV -16.8 percent to +12.6 percent, and MER -16.7 percent to +11.8 percent. These ranges replace the stale vendor artifacts that previously showed AEV touching -47.2 percent and MER +28.7 percent. The global assets display broader boxes and deeper tails: BTC spans -46.5 percent to +19.2 percent, META -30.6 percent to +20.9 percent, NVDA -20.4 percent to +21.8 percent, and SPY -11.6 percent to +10.0 percent. Domestic equities offer day-to-day stability, while the international technology and crypto positions carry the tail-event exposure that active risk management must address.

3.5 Cumulative Wealth Growth Trajectories

Figure 5 shows the cumulative wealth index of a hypothetical USD 1.00 investment from January 2020 through the end of the sample on a logarithmic scale. No-trade days are held at the last observed mark, so the terminal values equal the actual USD price ratios.

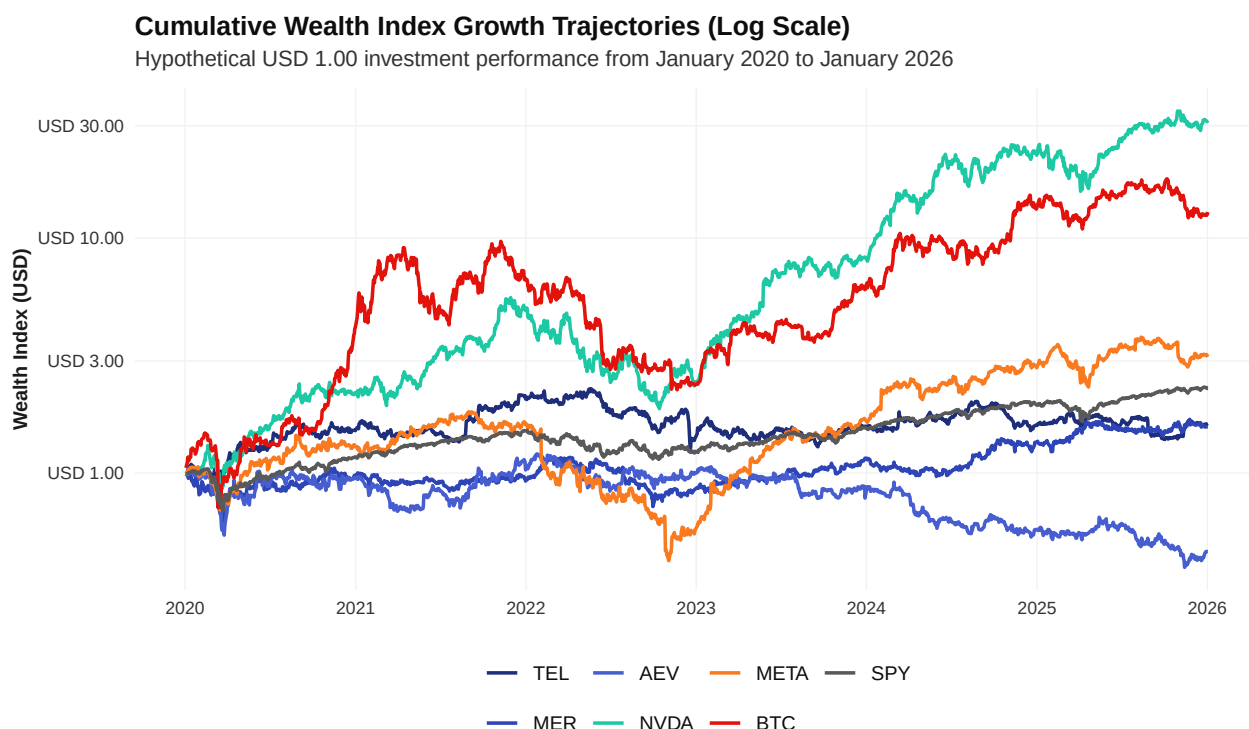


Figure 5: Cumulative Wealth Index Growth Trajectories (Log Scale)

The logarithmic plot reveals massive long-run divergence across the universe. NVIDIA compounded USD 1.00 into roughly USD 31.23 by the end of the sample, a total return of 3,023.4 percent, driven by the global expansion of artificial intelligence infrastructure demand. Bitcoin reached approximately USD 12.53 (1,152.7 percent) despite a drawdown of more than 60 percent during the 2022 crypto winter, while META reached USD 3.17 (216.9 percent) and SPY USD 2.29 (129.1 percent). The Philippine block delivered modest positive outcomes for TEL and MER, ending at USD 1.62 (61.7 percent) and USD 1.57 (57.2 percent) respectively, the latter recovering from three down years in 2020–2022 to compound strongly through 2023–2025. AEV was the sole structural loser, ending at USD 0.46 (-53.6 percent) after consecutive calendar-year

declines of -21.6 percent (2023), -26.5 percent (2024), and -19.3 percent (2025), with calendar-year returns measured on returns dated within each year. The divergence demonstrates that high-volatility global assets generated substantial long-run wealth, while the domestic names acted as stability anchors rather than growth engines.

3.6 Asset Return Correlation Structure and Diversification Potential

Table 3 presents the pairwise return correlation matrix computed on observed returns (pairwise complete observations), and Figure 6 displays the corresponding heatmap.

Table 3: Pairwise Asset Return Correlation Matrix (Jan 2, 2020 – Dec 31, 2025)

Asset	TEL	MER	AEV	NVDA	META	BTC	SPY
TEL	1.0000	0.1187	0.0884	0.1894	0.1647	0.1615	0.3010
MER	0.1187	1.0000	0.2959	0.0121	0.0172	0.0286	0.0518
AEV	0.0884	0.2959	1.0000	0.0155	0.0356	0.0155	0.0904
NVDA	0.1894	0.0121	0.0155	1.0000	0.5333	0.3204	0.7052
META	0.1647	0.0172	0.0356	0.5333	1.0000	0.2438	0.6285
BTC	0.1615	0.0286	0.0155	0.3204	0.2438	1.0000	0.3880
SPY	0.3010	0.0518	0.0904	0.7052	0.6285	0.3880	1.0000

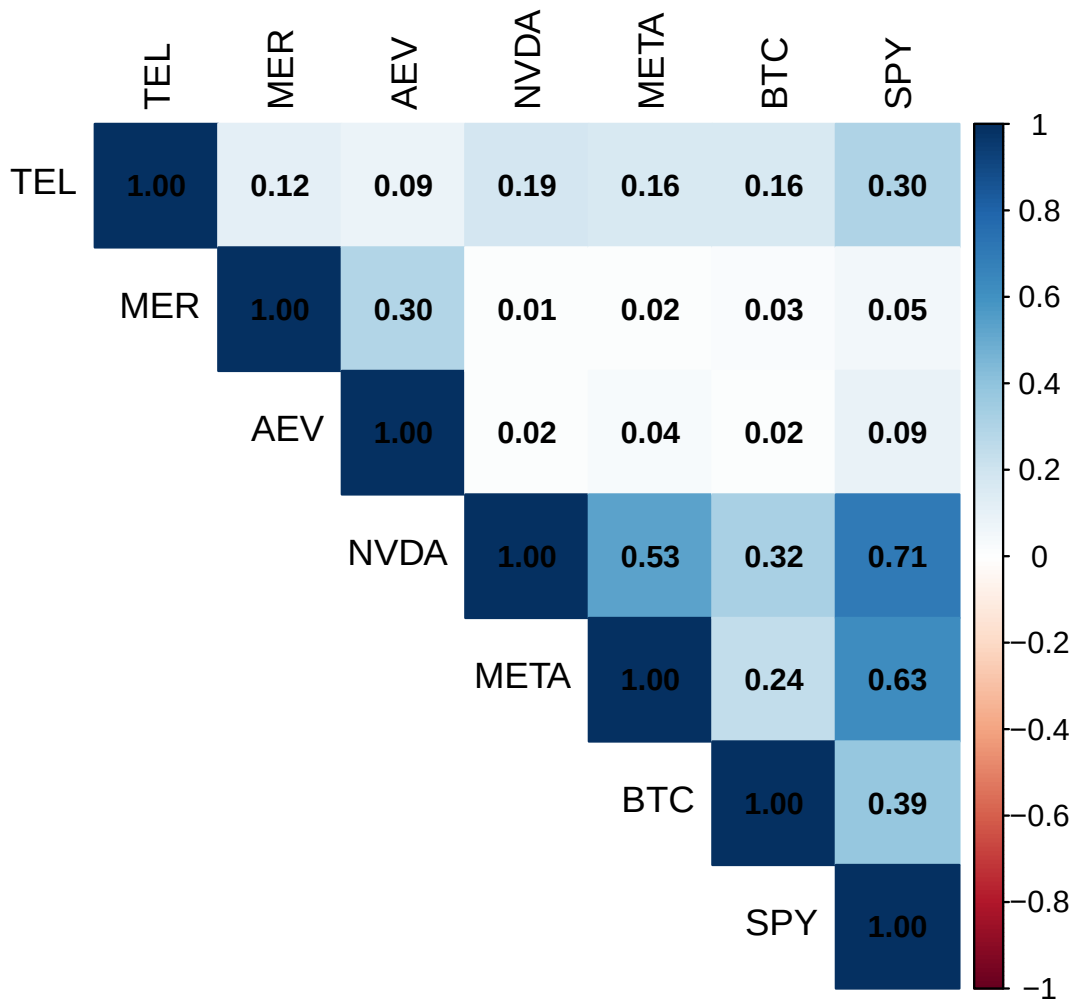


Figure 6: Asset Return Correlation Heatmap (Group 3 Universe)

The correlation structure is now estimated on genuinely traded PSE returns, and it paints a materially different picture from the vendor-sourced data. Within the global block, co-movement is strong: NVDA and META correlate at +0.5333, NVDA with SPY at +0.7052, META with SPY at +0.6285, and BTC with SPY at +0.3880. These linkages reflect common exposure to global liquidity and risk sentiment.

The Philippine block displays moderate positive internal co-movement: MER-TEL at +0.1187, AEV-TEL at +0.0884, and MER-AEV at +0.2959. The earlier claim of negative domestic pairwise correlations was an artifact of stale zero-volume OTC quotes and is not supported by the PSE data. What the corrected data do support is a genuine cross-market diversification story: the domestic names are nearly orthogonal to global risk assets. MER correlates +0.0121

with NVDA, +0.0172 with META, +0.0286 with BTC, and +0.0518 with SPY; AEV records +0.0155, +0.0356, +0.0155, and +0.0904 against the same four series. TEL, whose NYSE-listed ADR (PHI) trades in US hours, shows the strongest international linkage of the domestic set, at +0.3010 with SPY and +0.1894 with NVDA.

These near-zero cross-market coefficients provide the quantitative justification for combining high-return international technology and crypto assets with Philippine equities. The diversification benefit is real and now rests on clean data: domestic positions add variance-reduction potential with essentially no correlation drag from global tech and crypto shocks, while the moderate internal correlations among TEL, MER, and AEV preserve a degree of local systematic risk that sectoral selection can further diversify.

Overall, the exploratory financial analysis reveals a clear distinction between the portfolio's domestic and global assets. Philippine equities generally exhibit lower volatility and weak correlations with global technology and cryptocurrency assets, supporting their role as a diversification and stability anchors. Conversely, NVDA, META, and BTC provide substantially higher return potential but also exhibit greater volatility and exposure to extreme losses. These findings establish the fundamental risk-return trade-offs that guide the portfolio construction and optimization strategies examined in the succeeding sections.

4 Portfolio Construction and Strategy Evaluation

Portfolio construction transforms individual asset return dynamics into integrated allocation strategies. This section evaluates the performance of equal-weighted multi-asset portfolios under two execution regimes: an un-rebalanced Buy-and-Hold strategy and a Monthly Rebalanced strategy. Both start from an equal allocation vector ($w_i = 1/6 \approx 16.67$ percent) across the six-asset universe (TEL, MER, AEV, NVDA, META, BTC) on January 2, 2020 and are benchmarked against the SPDR S&P 500 ETF Trust (SPY). Portfolio arithmetic uses a carried price series at the aggregation layer: days when the PSE is closed contribute no return, and resumption days capture the full move across the gap. The stored master dataset itself contains no imputed rows, as documented in the data acquisition chapter. Portfolio returns are computed on simple (arithmetic) daily returns and compounded directly, so terminal wealth equals the value of the actual share-weighted holdings.

4.1 Comparative Strategy Performance Evaluation

Table 4 presents cumulative returns, annualized return (CAGR), annualized volatility, Sharpe ratios (using the 5.25 percent Philippine 91-day T-bill rate as the risk-free benchmark), and maximum drawdowns for both strategies and the SPY benchmark.

Table 4: Portfolio Performance Summary Comparison (Jan 2, 2020 – Dec 31, 2025)

Strategy	Cumulative	CAGR	Volatility	Sharpe	MaxDD
Equal-Weight (Buy & Hold)	743.04%	42.83%	35.12%	1.0700	58.21%
Equal-Weight (Monthly Rebalanced)	384.85%	30.21%	24.18%	1.0325	43.20%
SPDR S&P 500 ETF (SPY Benchmark)	129.06%	14.87%	20.75%	0.4635	33.72%

Both portfolio configurations generated substantial alpha relative to the SPY benchmark. The Buy-and-Hold strategy delivered a cumulative return of 743.04 percent and a 42.83 percent CAGR, nearly six times the benchmark’s cumulative return. This outperformance was driven by unconstrained weight compounding in NVIDIA and Bitcoin, which together absorbed roughly 86.5 percent of the portfolio’s capital by the end of the sample. The cost of that concentration is

visible in the risk metrics: annualized volatility of 35.12 percent and a maximum drawdown of -58.21 percent, the deepest in the table.

Monthly rebalancing re-engineered the risk-return trade-off. The disciplined portfolio returned 384.85 percent cumulatively (30.21 percent CAGR), surrendering 358.2 percentage points of cumulative return relative to Buy-and-Hold but cutting annualized volatility by 10.94 percentage points to 24.18 percent and trimming the maximum drawdown by 15.01 percentage points to -43.20 percent. On a risk-adjusted basis the two strategies are close, with the Buy-and-Hold Sharpe ratio of 1.0700 edging the rebalanced 1.0325; over this particular sample the foregone compounding outweighed the risk saved. Both Sharpe ratios, and both portfolios' absolute returns, remain far ahead of the benchmark's 0.4635 and 129.06 percent.

4.2 Cumulative Wealth Growth Dynamics

Figure 7 compares the cumulative wealth trajectories of a USD 1.00 initial allocation across the two strategies and SPY on a logarithmic scale.

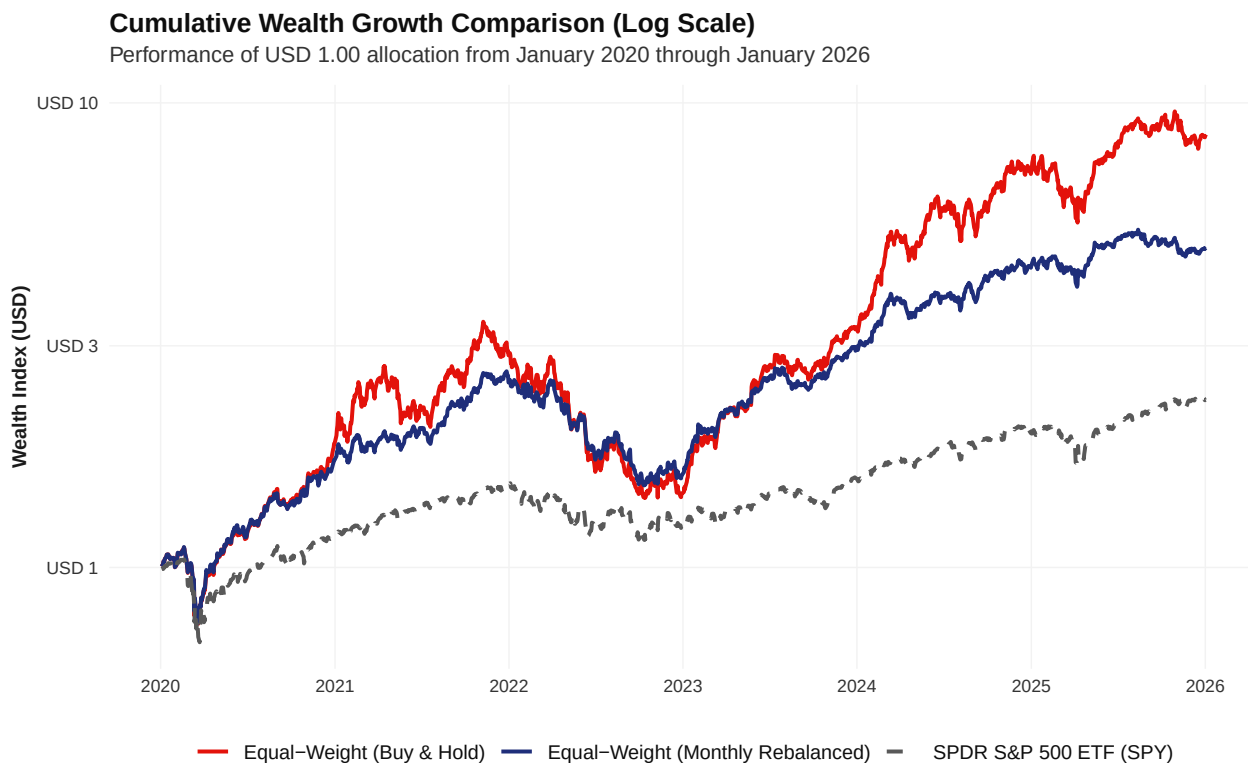


Figure 7: Cumulative Wealth Growth Comparison (Log Scale)

The trajectories terminate at USD 8.43 (Buy-and-Hold), USD 4.85 (Monthly Rebalanced), and USD 2.29 (SPY). The divergence emerged early: calendar-year 2020 returns were 84.72 percent for Buy-and-Hold, 70.37 percent for the rebalanced portfolio, and 17.24 percent for SPY, followed by 2021 gains of 54.93 percent, 46.80 percent, and 28.73 percent respectively. The rebalanced portfolio systematically realized profits from the surging technology and crypto positions at each monthly reset and redeployed the proceeds into the lagging domestic names, which kept its wealth path smoother but also capped its participation in the 2020–2021 bull market.

The 2022 tightening cycle was the stress test that separated the regimes. Buy-and-Hold lost 49.57 percent in calendar 2022 and carved out its maximum drawdown of -58.21 percent from its November 8, 2021 peak to a trough on October 14, 2022, as NVIDIA (-50.3 percent), Meta (-64.2 percent), and Bitcoin (-64.2 percent) collapsed simultaneously while even TEL fell 31.7 percent. The rebalanced portfolio absorbed the same shock with a 36.77 percent calendar-year loss and a -43.20 percent drawdown that bottomed on October 12, 2022, from a peak on December 27, 2021. Monthly profit-taking and redeployment into the comparatively resilient Philippine names (MER -7.6 percent, AEV -3.3 percent in 2022) therefore did not shorten the drawdown, which the 2023 recovery resolved for both strategies at roughly the same time; what rebalancing bought was depth, shaving 15.01 percentage points off the trough. SPY, by contrast, lost 18.18 percent in 2022 with a peak-to-trough drawdown of -33.72 percent confined to the March 2020 COVID crash.

The recovery phase reinforced the same pattern. In 2023 the Buy-and-Hold portfolio returned 125.47 percent and in 2024 a further 116.05 percent, powered by the AI-driven re-rating of NVIDIA and the crypto recovery; the rebalanced portfolio returned 85.09 percent and 50.88 percent over the same two years, and SPY 26.18 percent and 24.89 percent. In 2025 the rebalanced portfolio's diversified exposure lagged, returning 9.78 percent against 19.92 percent for Buy-and-Hold and 17.72 percent for SPY, as the domestic block and Bitcoin detracted. Over the full window, disciplined rebalancing traded roughly 1.74x of terminal wealth for a materially calmer path.

4.3 Asset Weight Drift and Allocation Evolution

Figure 8 displays the evolution of asset weight shares in the un-rebalanced Buy-and-Hold portfolio.

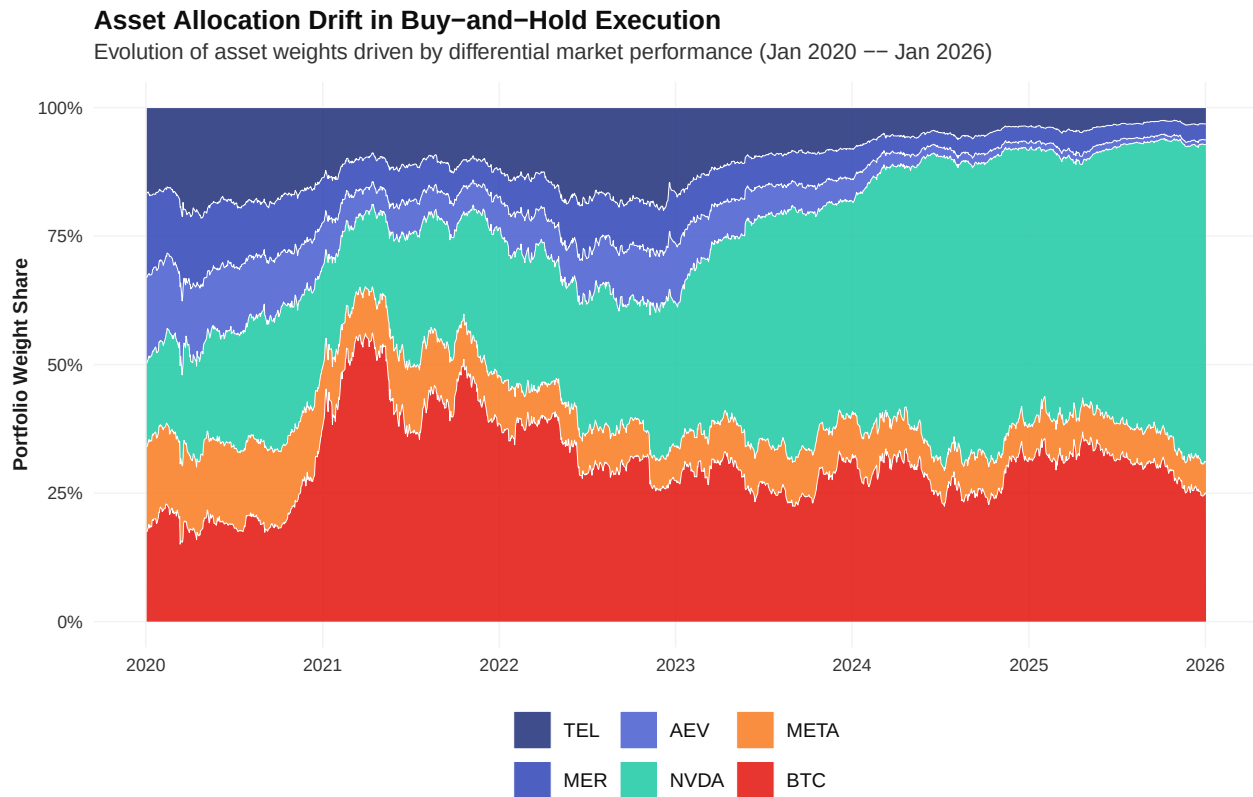


Figure 8: Asset Weight Drift Over Time in the Un-Rebalanced Buy-and-Hold Portfolio

The drift visualizer exposes the core vulnerability of passive execution. Each asset began at a 16.67 percent weight on January 2, 2020. Differential performance then restructured the allocation completely: by the end of the sample, NVIDIA had grown to 61.7 percent and Bitcoin to 24.8 percent, with META at 6.3 percent, while the three Philippine equities were diluted to 3.2 percent (TEL), 3.1 percent (MER), and 0.9 percent (AEV). The technology-cryptocurrency pair finished at 86.5 percent of the portfolio, and NVIDIA's weight peaked at 62.3 percent in late December 2025. The Buy-and-Hold portfolio thus ended the period as a concentrated two-asset bet with a small domestic hedge, a transformation that explains both its extraordinary cumulative return and its deep drawdown.

Monthly rebalancing prevents this drift by construction: on the first trading day of each month the weights are restored to 16.67 percent, selling the appreciated technology and crypto positions and buying the underweight domestic and lagging assets. This mechanical discipline enforces a systematic contrarian stance that caps concentration risk and keeps the portfolio aligned with its original diversification intent, at the measured cost of lower realized cumulative return over a sample dominated by two assets' exceptional appreciation.

5 Portfolio Optimization

Mean-variance optimization establishes the mathematical foundation for modern asset allocation by identifying the locus of portfolios that maximize expected return for a defined level of variance, or equivalently, minimize variance for a target expected return. This section formulates the optimization framework, derives the analytical and numerical solutions for the Global Minimum Variance (GMV) and Maximum Sharpe Ratio (Tangency) portfolios, and provides an extensive econometric interpretation of the optimal asset weightings and the Efficient Frontier.

5.1 Mathematical Framework and Convex Optimization Formulation

Let the asset universe consist of $n = 6$ constituents with annualized expected return vector $\boldsymbol{\mu} \in \mathbb{R}^n$ and annualized symmetric positive semi-definite covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$. Portfolio allocations are represented by the weight vector $\mathbf{w} = [w_1, w_2, \dots, w_n]^T \in \mathbb{R}^n$. All allocations are restricted to the unit simplex $\Delta^{n-1} = \{\mathbf{w} \in \mathbb{R}^n : \mathbf{w}^T \mathbf{1} = 1, w_i \geq 0 \forall i\}$, enforcing full investment and strictly prohibiting short sales.

5.1.1 Global Minimum Variance Portfolio Formulation

The Global Minimum Variance portfolio represents the leftmost boundary point of the Markowitz bullet, minimizing total portfolio variance without regard to expected return. Formally, the primal quadratic program is defined as:

$$\min_{\mathbf{w} \in \mathbb{R}^n} \quad \frac{1}{2} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{1} = 1, \quad w_i \geq 0 \quad \forall i \in \{1, \dots, n\} \quad (5)$$

In the unconstrained benchmark case that permits short sales, the Lagrangian function is given by $\mathcal{L}(\mathbf{w}, \lambda) = \frac{1}{2} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} - \lambda(\mathbf{w}^T \mathbf{1} - 1)$. Setting first-order stationarity conditions $\nabla_{\mathbf{w}} \mathcal{L} = \boldsymbol{\Sigma} \mathbf{w} - \lambda \mathbf{1} = \mathbf{0}$ yields the classic analytical solution:

$$\mathbf{w}_{\text{GMV, unconstrained}}^* = \frac{\boldsymbol{\Sigma}^{-1} \mathbf{1}}{\mathbf{1}^T \boldsymbol{\Sigma}^{-1} \mathbf{1}}, \quad \sigma_{\text{GMV}}^2 = \frac{1}{\mathbf{1}^T \boldsymbol{\Sigma}^{-1} \mathbf{1}} \quad (6)$$

Under the long-only constraint ($w_i \geq 0$), the Karush-Kuhn-Tucker first-order optimality conditions require:

$$\Sigma \mathbf{w}^* - \lambda \mathbf{1} - \boldsymbol{\nu}^* = \mathbf{0}, \quad \mathbf{w}^* \geq \mathbf{0}, \quad \mathbf{1}^T \mathbf{w}^* = 1, \quad \boldsymbol{\nu}^* \geq \mathbf{0}, \quad \nu_i^* w_i^* = 0 \quad \forall i \quad (7)$$

where $\lambda \in \mathbb{R}$ is the Lagrange multiplier associated with the budget constraint, and $\boldsymbol{\nu}^* \in \mathbb{R}_+^n$ is the vector of dual Lagrange multipliers enforcing the non-negativity bounds. When an asset allocation is constrained to zero ($w_i^* = 0$), its corresponding dual multiplier $\nu_i^* > 0$ represents the shadow cost, measuring the marginal variance penalty that would be incurred if a non-zero allocation were forced upon the portfolio.

5.1.2 Maximum Sharpe Ratio Portfolio Formulation

The Tangency portfolio identifies the unique coordinate along the Efficient Frontier that maximizes excess return per unit of total risk relative to the Philippine 91-Day Treasury Bill baseline yield ($R_f = 5.25\%$ per annum):

$$\max_{\mathbf{w} \in \mathbb{R}^n} \frac{\mathbf{w}^T \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^T \Sigma \mathbf{w}}} \quad \text{subject to} \quad \mathbf{w}^T \mathbf{1} = 1, \quad w_i \geq 0 \quad \forall i \quad (8)$$

Because the Sharpe ratio objective is non-linear and fractional, it can be mapped into an equivalent strictly convex quadratic program (Sharpe, 1964). Applying the variable transformation $\mathbf{y} = \kappa \mathbf{w}$ with scaling factor $\kappa = (\mathbf{w}^T \boldsymbol{\mu} - R_f)^{-1} > 0$, the optimization maps to:

$$\min_{\mathbf{y} \in \mathbb{R}^n} \frac{1}{2} \mathbf{y}^T \Sigma \mathbf{y} \quad \text{subject to} \quad (\boldsymbol{\mu} - R_f \mathbf{1})^T \mathbf{y} = 1, \quad \mathbf{y} \geq \mathbf{0} \quad (9)$$

Upon solving for the optimal vector $\mathbf{y}^*$ via quadratic programming, the normalized portfolio weights are recovered through the relation $\mathbf{w}_{\text{Tangency}}^* = \mathbf{y}^* / (\mathbf{1}^T \mathbf{y}^*)$.

5.2 Empirical Allocation Weights and Economic Mechanisms

Table 5 details the numerical weights, annualized returns, annualized standard deviations, and Sharpe ratios for the Global Minimum Variance and Maximum Sharpe Ratio portfolios solved via the Sequential Least Squares Programming and Goldfarb-Idnani dual quadratic programming algorithms.

Table 5: Optimal Portfolio Weights: GMV vs Maximum Sharpe

Asset	GMV Weight	Max Sharpe Weight
TEL	31.01%	0.00%
MER	31.23%	18.42%
AEV	16.85%	0.00%
NVDA	3.15%	55.44%
META	12.86%	0.00%
BTC	4.90%	26.14%

The empirical weights in Table 5 demonstrate how mathematical optimization exploits cross-asset covariance structures to satisfy divergent investment objectives.

The Global Minimum Variance portfolio allocates 79.09 percent of its capital to domestic Philippine equities, allocating 31.01 percent to PLDT Inc. (TEL), 31.23 percent to Manila Electric Company (MER), and 16.85 percent to Aboitiz Equity Ventures (AEV). This concentration stems directly from the low standalone volatilities of Philippine utility and telecom cash flows, where annualized standard deviations remain bounded at 31.13 percent for TEL, 31.38 percent for MER, and 38.91 percent for AEV, measured as the square root of the annualized covariance-matrix diagonal used in the optimization (simple daily returns annualized on a 252-day convention). Under the Euler risk decomposition theorem, the marginal contribution to risk must be equalized across all active positions in the minimum-variance state. Consequently, the optimizer pairs these domestic assets to minimize diagonal variance terms while exploiting low inter-asset correlations ($r_{\text{TEL}, \text{MER}} = 0.1187$, $r_{\text{TEL}, \text{AEV}} = 0.0884$, $r_{\text{MER}, \text{AEV}} = 0.2959$).

A notable empirical finding is the 3.15 percent allocation assigned to NVIDIA Corporation (NVDA) within the GMV portfolio. Despite NVIDIA exhibiting a high standalone volatility of 53.18 percent, its pairwise correlation with domestic Philippine equities is exceptionally low,

registering at 0.1894 with TEL and 0.0121 with MER. In matrix terms, the marginal risk derivative $\partial\sigma_p/\partial w_i = (\Sigma\mathbf{w})_i/\sigma_p$ demonstrates that introducing an uncorrelated foreign asset dampens aggregate portfolio variance because the off-diagonal covariance savings outweigh the small diagonal variance contribution. The same mechanism admits META (12.86 percent) and BTC (4.90 percent) at modest weights: their correlations with the domestic block are near zero (MER–META 0.0172, MER–BTC 0.0286), offsetting their higher diagonal variances. The resulting GMV portfolio delivers an annualized expected return of 16.03 percent with an annualized volatility of 20.22 percent, achieving a Sharpe ratio of 0.5332.

In contrast, the Maximum Sharpe Ratio portfolio shifts capital aggressively into high-momentum growth engines. The tangency vector allocates 55.44 percent to NVIDIA Corporation (NVDA), 26.14 percent to Bitcoin (BTC), and 18.42 percent to Manila Electric Company (MER). The heavy concentration in NVDA and BTC reflects the compounding capital appreciation experienced by US large-capitalization technology and cryptocurrency assets during the sample period, which expands the expected excess return numerator ($\mathbf{w}^T\boldsymbol{\mu} - R_f$) at a rate that far outpaces the expansion of portfolio variance in the denominator.

Manila Electric Company (MER) serves as the sole domestic anchor within the Maximum Sharpe portfolio, capturing nearly a fifth of total allocation. MER combines a strong historical return trajectory with utility dividend stability and near-zero correlation with global technology equities ($r_{\text{MER}, \text{NVDA}} = 0.0121$, $r_{\text{MER}, \text{BTC}} = 0.0286$). By incorporating MER, the optimizer mitigates currency and equity volatility shocks without diluting overall return generation. Telecommunications (TEL) and industrial conglomerates (AEV) are excluded from the tangency portfolio because their lower expected returns (12.96 percent and -5.26 percent annualized) dilute the Sharpe numerator, and Meta Platforms (META) receives zero weight despite its 28.99 percent expected return: at 43.82 percent volatility and a 0.5333 correlation with NVDA, its risk-adjusted contribution lies below the tangency line defined by NVDA, BTC, and MER. The resulting Maximum Sharpe portfolio achieves an annualized expected return of 58.10 percent and an annualized volatility of 38.25 percent, achieving the highest Sharpe ratio among the optimized portfolios at 1.3815.

5.3 Geometry of the Efficient Frontier and Capital Allocation Line

Figure 9 illustrates the Markowitz Efficient Frontier generated across 50 target return increments, beginning at the GMV vertex and terminating at the maximum return boundary, with the GMV and Maximum Sharpe portfolios marked.



Figure 9: Efficient Frontier with Global Minimum Variance and Maximum Sharpe Portfolios

The geometric properties displayed in Figure 9 illustrate the core principles of portfolio efficiency. The efficient frontier hyperbola originates at the GMV coordinate ($\sigma = 20.22\%$, $\mu = 16.03\%$) and curves upward with diminishing marginal returns to incremental volatility, terminating at the maximum-return boundary. The Maximum Sharpe portfolio ($\sigma = 38.25\%$, $\mu = 58.10\%$) marks the tangency point of the risk-free rate line with the risky asset envelope: under the separation theorem, an investor can move along the line connecting the 5.25 percent risk-free rate to this portfolio through risk-free lending or borrowing, and the slope of that line equals the maximum Sharpe ratio (1.3815). The figure plots only the frontier and the two optimal portfolios; individual assets and the Capital Allocation Line are not shown.

Figure 10 provides a comparative bar plot displaying the asset weight breakdown between the Global Minimum Variance and Maximum Sharpe Ratio portfolios.

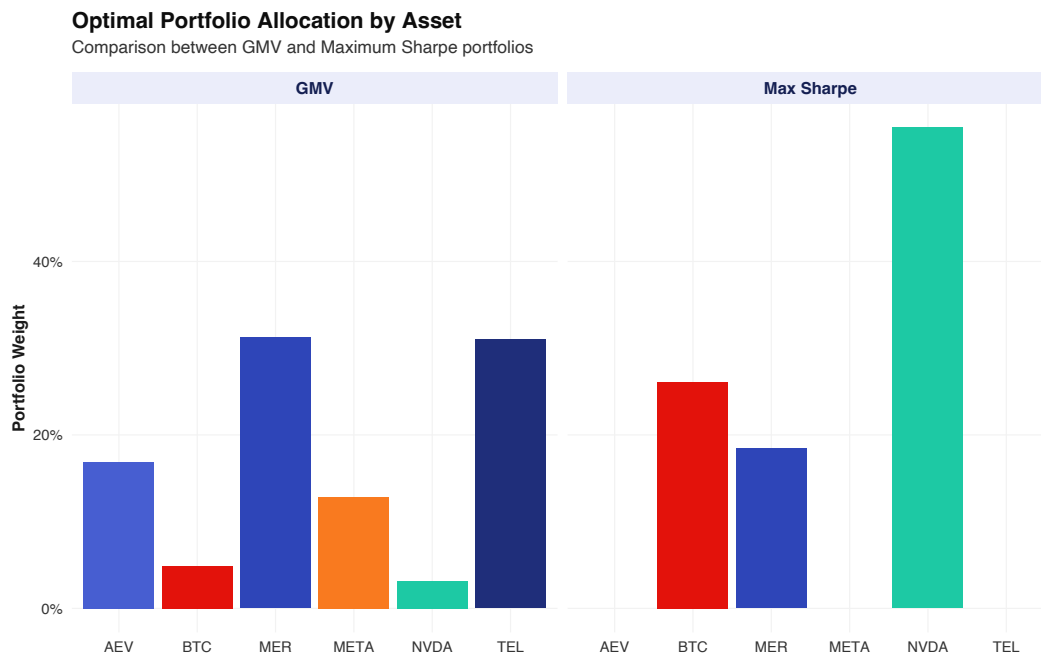


Figure 10: Optimal Asset Allocation Weights: Global Minimum Variance vs Maximum Sharpe Ratio

As illustrated in Figure 10, shifting investment objectives from absolute risk minimization to risk-adjusted return maximization induces a complete portfolio rotation. The defensive domestic base of TEL and AEV is entirely liquidated in favor of the highest-return global positions (NVDA and BTC), while MER provides the critical defensive ballast present in both investment mandates.

6 Portfolio Risk Assessment

While traditional mean-variance optimization relies on return variance as a symmetric measure of risk, empirical financial time series consistently violate Gaussian normality. Daily asset returns exhibit pronounced negative skewness, volatility clustering, and heavy tails (leptokurtosis). Consequently, relying solely on standard deviation significantly underestimates the probability and severity of catastrophic tail losses during periods of market turmoil.

To establish an institutional-grade risk evaluation, this section assesses downside vulnerability across the Global Minimum Variance (GMV) and Maximum Sharpe Ratio (Tangency) portfolios using non-parametric Value at Risk (VaR), coherent Expected Shortfall (CVaR), multivariate normal Monte Carlo simulations, and historical crisis stress testing.

6.1 Econometric Foundations of Downside Risk and Coherent Risk Measures

6.1.1 Value at Risk Formulation

Let $L = -\mathbf{w}^T \mathbf{r}$ denote the daily loss random variable of a portfolio with weight vector $\mathbf{w} \in \mathbb{R}^n$ and daily log return vector $\mathbf{r} \in \mathbb{R}^n$. The Value at Risk at confidence level $\alpha \in (0, 1)$ over a 1-day holding horizon is defined as the α -quantile of the portfolio loss distribution:

$$\text{VaR}_\alpha(\mathbf{w}) = \inf \{l \in \mathbb{R} : P(L > l) \leq 1 - \alpha\} = -F_R^{-1}(1 - \alpha) \quad (10)$$

where $F_R(r)$ is the cumulative distribution function of portfolio return $R = \mathbf{w}^T \mathbf{r}$. We compute two distinct specifications:

Non-parametric Historical VaR is calculated directly from the empirical quantile of observed sample returns without imposing distributional assumptions:

$$\text{VaR}_\alpha^{\text{Hist}}(\mathbf{w}) = -\widehat{F}_n^{-1}(1 - \alpha) = -R_{(\lfloor N(1-\alpha) \rfloor)} \quad (11)$$

where $R_{(k)}$ denotes the k -th order statistic across the $N = 1,507$ daily trading observations.

Parametric Gaussian VaR assumes that portfolio returns follow a stationary normal distribution $\mathbf{r} \sim \mathcal{N}(\boldsymbol{\mu}_{\text{daily}}, \boldsymbol{\Sigma}_{\text{daily}})$:

$$\text{VaR}_{\alpha}^{\text{Gauss}}(\mathbf{w}) = -\left(\mathbf{w}^T \boldsymbol{\mu}_{\text{daily}} + z_{1-\alpha} \sqrt{\mathbf{w}^T \boldsymbol{\Sigma}_{\text{daily}} \mathbf{w}}\right) \quad (12)$$

where $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$ is the standard normal inverse cumulative distribution value ($z_{0.05} \approx -1.6449$ for $\alpha = 0.95$, and $z_{0.01} \approx -2.3263$ for $\alpha = 0.99$).

6.1.2 Expected Shortfall and Coherent Risk Axioms

Although Value at Risk remains the standard benchmark in financial regulation, it suffers from two major theoretical flaws: it is completely silent regarding the magnitude of losses exceeding the quantile threshold, and it fails the mathematical property of subadditivity under non-normal return distributions. When subadditivity is violated, combining portfolios can produce an aggregate VaR greater than the sum of individual VaRs, penalizing diversification.

To address these limitations, Expected Shortfall (ES_{α} , or Conditional Value at Risk) measures the conditional expectation of loss given that the loss strictly exceeds the VaR_{α} threshold:

$$\text{ES}_{\alpha}(\mathbf{w}) = \mathbb{E}[L \mid L \geq \text{VaR}_{\alpha}(\mathbf{w})] = \frac{1}{1 - \alpha} \int_{\alpha}^1 \text{VaR}_u(\mathbf{w}) du \quad (13)$$

As proven by Artzner et al. (1999), Expected Shortfall constitutes a strictly coherent risk measure by satisfying four fundamental axioms: Translation Invariance, Subadditivity ($\text{ES}_{\alpha}(L_1 + L_2) \leq \text{ES}_{\alpha}(L_1) + \text{ES}_{\alpha}(L_2)$), Positive Homogeneity, and Monotonicity. Under Gaussian normality, Expected Shortfall is computed analytically following Fabozzi et al. (2002):

$$\text{ES}_{\alpha}^{\text{Gauss}}(\mathbf{w}) = -\mathbf{w}^T \boldsymbol{\mu}_{\text{daily}} + \sigma_p \frac{\phi(z_{1-\alpha})}{1 - \alpha} \quad (14)$$

where $\phi(\cdot)$ is the standard normal probability density function and $\sigma_p = \sqrt{\mathbf{w}^T \boldsymbol{\Sigma}_{\text{daily}} \mathbf{w}}$.

6.2 Comparative Empirical Tail Risk and Underestimation Bias

Table 6 reports the daily 95 percent and 99 percent Historical and Parametric VaR and Expected Shortfall metrics across the 1,507 daily trading observations for the GMV and Maximum Sharpe Ratio portfolios.

Table 6: Historical and Parametric VaR / CVaR Summary (Daily, 95 percent and 99 percent Confidence)

Portfolio	Metric	95% CI	99% CI
GMV	Historical VaR	1.80%	3.27%
GMV	Parametric VaR	2.03%	2.90%
GMV	Historical ES (CVaR)	2.85%	4.67%
GMV	Parametric ES (CVaR)	2.56%	3.33%
Max Sharpe	Historical VaR	3.39%	5.64%
Max Sharpe	Parametric VaR	3.74%	5.38%
Max Sharpe	Historical ES (CVaR)	5.11%	8.55%
Max Sharpe	Parametric ES (CVaR)	4.74%	6.19%

The empirical metrics in Table 6 reveal substantial divergence in downside vulnerability between the two portfolio allocations.

The Global Minimum Variance portfolio demonstrates exceptional downside containment. Over a daily horizon, its 95 percent Historical VaR is limited to 1.80 percent, with an Expected Shortfall of 2.85 percent. Under extreme 99 percent tail conditions, the GMV portfolio experiences a Historical VaR of 3.27 percent and an Expected Shortfall of 4.67 percent per day. Comparing the Historical metrics against Parametric Gaussian estimates ($VaR_{95} = 2.03\%$, $ES_{95} = 2.56\%$) confirms that the domestic utility and telecom allocation (TEL, AEV, MER) exhibits stable, near-Gaussian return behavior with minimal tail risk amplification.

Conversely, the Maximum Sharpe Ratio portfolio exhibits severe tail vulnerability driven by its 81.58 percent aggregate concentration in NVIDIA and Bitcoin (NVDA: 55.44 percent, BTC: 26.14 percent). Its daily 95 percent Historical VaR reaches 3.39 percent (nearly double that of the GMV portfolio), while its 95 percent Expected Shortfall expands to 5.11 percent. At the 99 percent confidence level, the Maximum Sharpe portfolio incurs a Historical VaR of 5.64 percent and an empirical Expected Shortfall of 8.55 percent per day.

Crucially, the empirical 99 percent Expected Shortfall (8.55 percent) exceeds the Gaussian parametric prediction (6.19 percent) by 236 basis points per day, representing a 38.1 percent relative underestimation error. This discrepancy demonstrates the presence of pronounced negative tail jumps in technology and cryptocurrency positions, proving that conventional Gaussian risk models dangerously understate actual capital requirements during severe market corrections.

6.3 Multivariate Monte Carlo Return Simulation and Density Geometry

To evaluate potential return trajectories beyond the observed historical sample, a 10,000-draw Monte Carlo simulation was executed using the multivariate normal covariance structure of the six assets. Synthetic return vectors $\mathbf{r}_{\text{sim}} \in \mathbb{R}^6$ were generated as $\mathbf{r}_{\text{sim}} = \boldsymbol{\mu}_{\text{daily}} + \mathbf{V}\boldsymbol{\Lambda}^{1/2}\mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_6)$, where $\boldsymbol{\Sigma}_{\text{daily}} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T$ is the eigen decomposition of the daily covariance matrix used by the `MASS::mvrnorm` implementation.

Table 7 reports the simulated Value at Risk and Expected Shortfall metrics across the 10,000 synthetic trading days.

Table 7: Monte Carlo Simulated VaR and Expected Shortfall (10,000 Simulations)

Portfolio	MC VaR (95%)	MC ES (95%)
GMV	2.02%	2.54%
Max Sharpe	3.68%	4.72%

The Monte Carlo simulation produces a 95 percent MC VaR of 2.02 percent and a 95 percent MC ES of 2.54 percent for the GMV portfolio. For the Maximum Sharpe portfolio, the simulated 95 percent MC VaR reaches 3.68 percent with an Expected Shortfall of 4.72 percent.

Figure 11 displays the kernel density overlay comparing the empirical historical return distribution against the 10,000-draw Monte Carlo simulated distribution for the Maximum Sharpe portfolio.

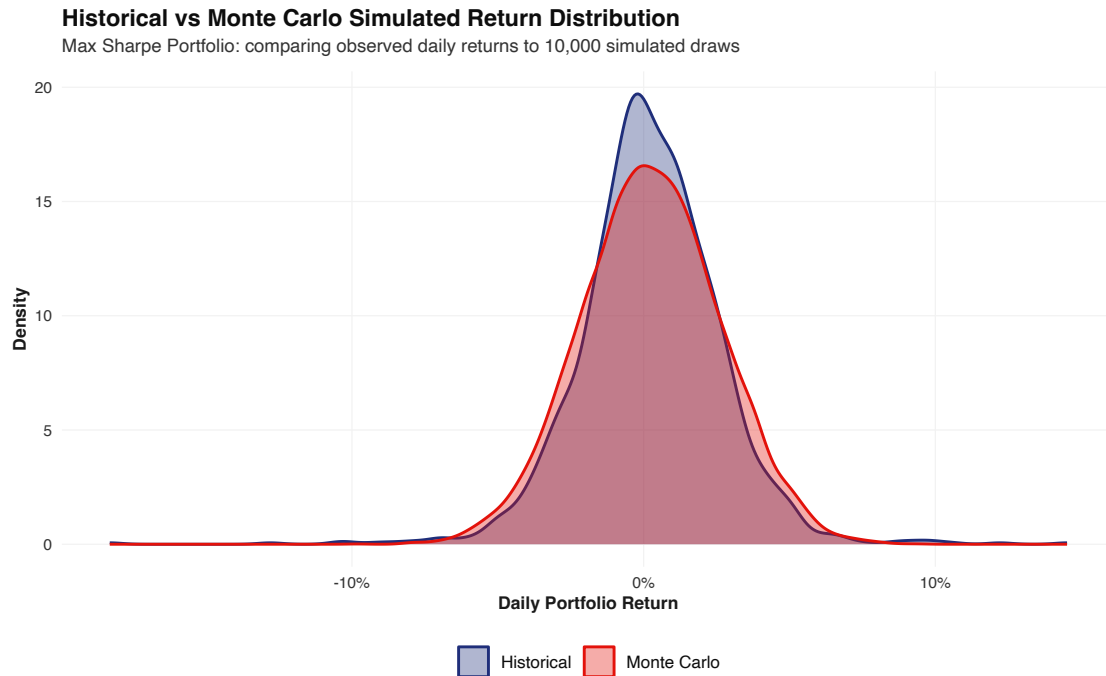


Figure 11: Historical vs 10,000-Draw Monte Carlo Simulated Return Distribution (Max Sharpe Portfolio)

The density overlay in Figure 11 provides visual confirmation of structural non-normality. The empirical return series (blue density) exhibits a significantly sharper, leptokurtic central peak relative to the Gaussian Monte Carlo benchmark (red density), reflecting the tendency of returns to cluster around the mean during calm market regimes. Simultaneously, the empirical left tail extends far beyond the Gaussian boundaries, illustrating that severe negative outliers occur with substantially greater frequency than predicted by normal distribution theory.

6.4 Macroeconomic Crisis Stress Testing and Capital Preservation Dynamics

To test structural solvency during severe macroeconomic shocks, portfolio capital draw-downs were simulated across five historical and hypothetical crisis scenarios: the 2020 COVID-19 Liquidity Shock (Feb 19 – Mar 23, 2020), the 2022 Crypto Winter and Stablecoin Contagion (May 5 – May 13, 2022), the 2022 Federal Reserve Aggressive Rate Hike Shock (Jun 8 – Jun 16,

2022), the Tech Sector Multiple De-rating (Nov 19, 2021 – Jan 27, 2022), and a Hypothetical 1.5x Multiplier Black Swan Shock.

Table 8 details the simulated portfolio drawdowns across all five crisis stress events.

Table 8: Portfolio Impact Under Stress Scenarios

Scenario	GMV Portfolio Impact	Max Sharpe Portfolio Impact
2020 COVID Crash	-22.55%	-28.83%
2022 Crypto Winter	-4.90%	-14.40%
Fed Rate Hike Shock	-9.47%	-18.65%
Tech Sector Selloff	3.71%	-24.58%
Hypothetical Black Swan	-35.89%	-45.30%

Figure 12 illustrates the comparative drawdown impacts between the GMV and Maximum Sharpe portfolios across all five crisis scenarios.

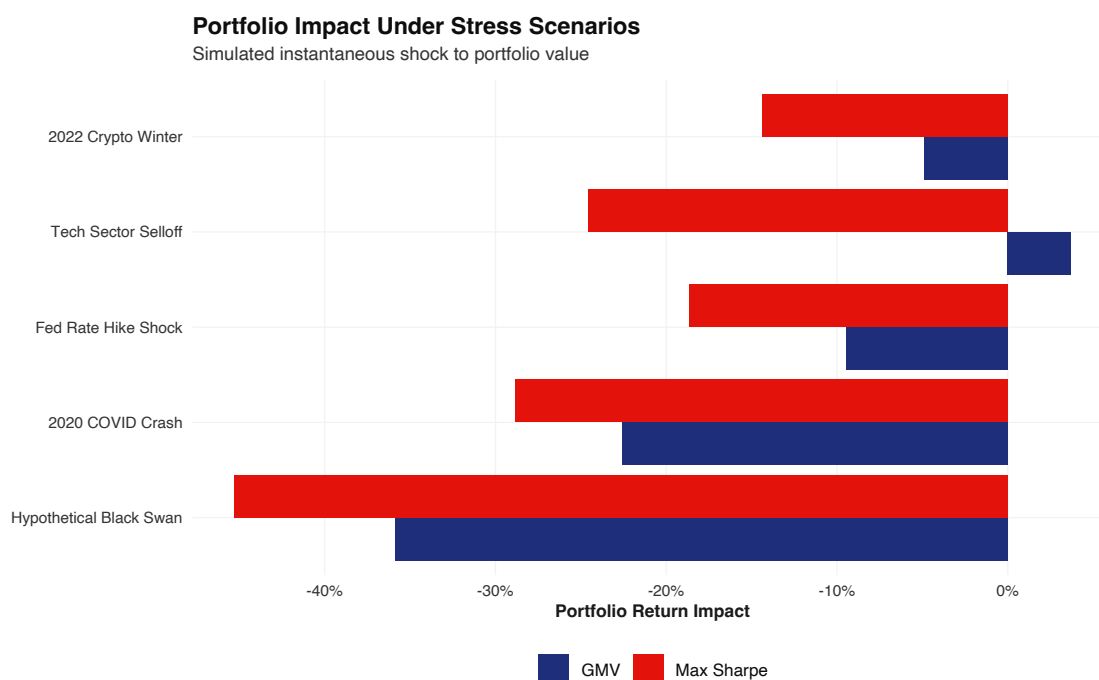


Figure 12: Portfolio Value Impact Across Crisis Stress Scenarios: GMV vs Maximum Sharpe Ratio

A granular economic analysis of Table 8 and Figure 12 demonstrates the capital preservation properties of the GMV allocation during market dislocations. During the 2020 COVID-19 pandemic crash, the GMV portfolio limited capital loss to -22.55 percent, whereas the Maximum Sharpe portfolio suffered a steep -28.83 percent drawdown (1.28 times greater capital loss) due to

synchronized global equity and crypto selling. During the 2022 Crypto Winter, the GMV portfolio declined by -4.90 percent, while the Maximum Sharpe portfolio lost -14.40 percent.

The 2022 Federal Reserve rate tightening shock further highlights the equity duration risk embedded in high-growth technology equities. Because growth equities derive a substantial portion of their valuation from distant future cash flows, rising discount rates compressed their present values, inflicting a -18.65 percent drawdown on the Maximum Sharpe portfolio compared to -9.47 percent for GMV. Similarly, during the broader tech valuation compression, the Maximum Sharpe portfolio fell -24.58 percent, while the GMV portfolio gained 3.71 percent. Under the hypothetical 1.5x Black Swan shock, the GMV drawdown was contained at -35.89 percent, whereas the Maximum Sharpe portfolio suffered a severe -45.30 percent capital loss.

These findings highlight the core trade-off governing quantitative portfolio construction. While the Maximum Sharpe allocation delivers superior wealth compounding during sustained economic expansions, the Global Minimum Variance allocation provides indispensable structural protection against systemic financial crises.

7 Executive Investment Recommendation

Based on the portfolio optimization, historical performance, downside-risk measures, and stress-testing results, the Maximum Sharpe Ratio portfolio is recommended. The portfolio allocates 55.44 percent to NVIDIA (NVDA), 26.14 percent to Bitcoin (BTC), and 18.42 percent to Manila Electric Company (MER), delivering an expected annual return of 58.10 percent, an annualized volatility of 38.25 percent, and a Sharpe ratio of 1.3815 at the 5.25 percent risk-free rate, the highest risk-adjusted profile in the feasible universe. Figure 10 contrasts this allocation with the GMV portfolio, and Figure 9 locates it at the tangency point of the efficient frontier.

NVIDIA contributes the strongest sample-period expected return (71.65 percent annualized) and, together with Bitcoin (61.49 percent annualized), drives the portfolio's return profile. Bitcoin's inclusion rests on its high expected return and its low correlation with the Philippine constituents (MER–BTC 0.0286), while NVIDIA pairs with the domestic block at near-zero correlation (MER–NVDA 0.0121, MER–META 0.0172) and at moderate co-movement with Bitcoin (NVDA–BTC 0.3204). Manila Electric Company anchors the allocation as the sole domestic constituent, combining a positive expected return (12.49 percent annualized) with near-zero correlation to both growth assets. TEL, AEV, and META receive zero weight in the long-only tangency solution.

Downside risk is governed by the empirical tail measures reported in Table 6. The recommended portfolio's daily 95 percent Historical VaR is -3.39 percent and its 95 percent Historical Expected Shortfall is -5.11 percent; at the 99 percent level these reach -5.64 percent and -8.55 percent. The empirical 99 percent Expected Shortfall (8.55 percent) exceeds the Gaussian parametric estimate (6.19 percent) by 236 basis points per day, so historical measures, rather than the normal approximation, should set the risk budget.

Stress testing confirms the trade-off between return and capital preservation. Under the five scenarios in Figure 12 and Table 8, the Maximum Sharpe portfolio loses 28.83 percent in the 2020 COVID crash, 24.58 percent in the technology-sector selloff, 18.65 percent in the 2022

rate-hike shock, and 14.40 percent in the 2022 crypto winter, with a -45.30 percent loss under the hypothetical 1.5x Black Swan scenario. The GMV portfolio contains the same shocks at materially smaller magnitudes (-22.55 percent COVID, -35.89 percent Black Swan), illustrating its role as the defensive alternative.

Disciplined monthly rebalancing is required to manage the portfolio's concentration: NVDA and BTC together represent 81.58 percent of the recommended allocation, and un-rebalanced equal-weight exposure drifted to 61.75 percent NVDA and 24.77 percent BTC by December 2025. Over the sample, monthly rebalancing reduced volatility from 35.12 percent to 24.18 percent and the peak drawdown from -58.21 percent to -43.20 percent at the cost of cumulative return (384.85 percent versus 743.04 percent). For investors prioritizing capital preservation, the GMV portfolio allocates TEL 31.01 percent, MER 31.23 percent, AEV 16.85 percent, NVDA 3.15 percent, META 12.86 percent, and BTC 4.90 percent, delivering a 16.03 percent expected return at 20.22 percent volatility (Sharpe 0.5332). Portfolio choice ultimately reflects the investor's willingness and capacity to bear losses: the Maximum Sharpe portfolio suits investors seeking higher returns despite greater risk, while the GMV portfolio suits investors who cannot tolerate peak stress losses near -45 percent.

8 Conclusion

This consultation constructed, optimized, and stress-tested a six-asset portfolio for Group 3 (Charlie)—TEL, MER, AEV, NVDA, META, and BTC—over the sample period January 2, 2020 through December 31, 2025, benchmarked against SPY at a 5.25 percent Philippine 91-day Treasury Bill risk-free rate.

The equal-weighted Buy-and-Hold strategy compounded USD 1.00 into USD 8.43 (743.04 percent cumulative, 42.83 percent CAGR) but ended as a concentrated two-asset bet, with NVDA and BTC reaching 61.75 percent and 24.77 percent of the portfolio and a -58.21 percent peak drawdown. Monthly rebalancing delivered USD 4.85 (384.85 percent, 30.21 percent CAGR) with materially lower volatility (24.18 percent versus 35.12 percent) and a shallower drawdown (-43.20 percent), demonstrating the discipline's risk-control value at the cost of some upside.

Mean-variance optimization formalized the trade-off. The GMV portfolio (TEL 31.01 percent, MER 31.23 percent, AEV 16.85 percent, NVDA 3.15 percent, META 12.86 percent, BTC 4.90 percent) achieves 16.03 percent return at 20.22 percent volatility (Sharpe 0.5332). The Maximum Sharpe portfolio (NVDA 55.44 percent, BTC 26.14 percent, MER 18.42 percent) achieves 58.10 percent return at 38.25 percent volatility and a Sharpe ratio of 1.3815, verified across multiple solver starts and cross-checked against a quadratic-programming frontier sweep.

Risk assessment showed that tail risk is real and non-Gaussian. The Maximum Sharpe portfolio's daily 99 percent Historical Expected Shortfall of 8.55 percent exceeds its Gaussian parametric estimate by 236 basis points, and its worst historical stress loss reaches -28.83 percent (COVID) with -45.30 percent under the 1.5x Black Swan scenario. The GMV portfolio generally lowers stress losses relative to the Maximum Sharpe portfolio, though the reduction is scenario-dependent.

The recommendation follows from these results: adopt the Maximum Sharpe portfolio under monthly rebalancing for investors with high loss capacity, or the GMV portfolio for capital

preservation, with historical tail-risk measures governing the risk budget. All quantitative claims in this report are reproducible from the canonical R pipeline (QMD) and the frozen numerical summary, and every figure and table reflects the final analytical state.

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Appendix A: Analytical Pipeline and Full Results

Portfolio Management and Risk Consultation

FINSTAD Case 2 – Group 3 (Charlie)

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1 Setup and Library Loading

Load the quantitative finance, portfolio analytics, and data visualization libraries required for the analysis.

```
# Core data manipulation and visualization
library(tidyverse)
library(lubridate)
library(scales)

# Time series and quantitative financial analysis
library(xts)
library(zoo)
library(PerformanceAnalytics)

# Solvers for portfolio optimization
library(nloptr)
library(quadprog)

# For Portfolio Risk Assessment
library(MASS)

# Table formatting and visual aesthetics
library(corrplot)
library(knitr)
library(kableExtra)
```

2 Executive Summary

This quantitative consultation report evaluates portfolio construction, optimization, and risk management strategies for Group 3 (Charlie) over the six-year sample period from January 2, 2020 through December 31, 2025. The mandated multi-asset investment universe comprises three Philippine equities (**TEL**, **MER**, **AEV**), two US international tech equities (**NVDA**, **META**), and one cryptocurrency (**BTC**), benchmarked against the S&P 500 ETF Trust (**SPY**) and evaluated against a 5.25% risk-free rate (Philippine 91-Day Treasury Bill).

2.0.1 Key Empirical Findings

1. **Portfolio Construction & Rebalancing Discipline:** An un-rebalanced Buy-and-Hold equal-weighted strategy delivered a cumulative return of 743.04% (42.83% CAGR) driven by massive asset weight drift into NVDA (61.75% of the portfolio by December 2025) and BTC (24.77%). Monthly rebalancing cut annualized volatility from 35.12% to 24.18% and contained the peak drawdown from -58.21% to -43.20%, at the cost of lower cumulative return (384.85%, 30.21% CAGR); the Sharpe ratio was marginally lower under rebalancing (1.0325 versus 1.0700).
2. **Mean-Variance Portfolio Optimization:**
 - **Global Minimum Variance (GMV):** Allocates 79.09% of capital to domestic Philippine equities (TEL: 31.01%, MER: 31.23%, AEV: 16.85%) with modest international tech exposure (NVDA: 3.15%, META: 12.86%) and 4.90% in BTC. GMV achieves an annualized return of **16.03%**, volatility of **20.22%**, and a Sharpe ratio of **0.5332**.
 - **Maximum Sharpe Ratio (Tangency Portfolio):** Concentrates in high-performing international growth engines (NVDA: 55.44%, BTC: 26.14%) plus MER (18.42%), with zero weights in TEL, AEV, and META, yielding an annualized return of **58.10%**, volatility of **38.25%**, and a Sharpe ratio of **1.3815**.
3. **Tail Risk & Stress Testing Resilience:**
 - Daily 95% Historical Value at Risk (VaR) is **-1.80%** (GMV) and **-3.39%** (Max Sharpe), while 95% Historical Expected Shortfall (CVaR) reaches **-2.85%** (GMV) and **-5.11%** (Max Sharpe).
 - Under historical crisis simulations (2020 COVID Crash, 2022 Crypto Winter, 2022 Fed Rate Hike), the GMV portfolio demonstrates resilience with a maximum historical shock loss of -22.55%, whereas the Max Sharpe portfolio incurs severe instantaneous drawdowns (-28.83% in the COVID crash, -45.30% in a 1.5x Black Swan event).

2.0.2 Executive Recommendation

We recommend the **Maximum Sharpe portfolio** (NVDA: 55.44%, BTC: 26.14%, MER: 18.42%) for investors with a high tolerance for volatility and drawdown, executed with **monthly systematic rebalancing**. With an expected return of **58.10%**, volatility of **38.25%**, and a Sharpe ratio of **1.3815** at a 5.25% risk-free rate, it is the most attractive risk-adjusted allocation in the universe. For investors prioritizing capital preservation, the **GMV portfolio** (TEL: 31.01%, MER: 31.23%, AEV: 16.85%, NVDA: 3.15%, META: 12.86%, BTC: 4.90%) is the defensive alternative, with 20.22% volatility, a daily 95% Historical VaR of -1.80%, and a maximum historical stress loss of -22.55%.

3 Introduction

The 2020–2025 sample period presented unprecedented structural shifts across global asset markets. Investors confronted the shock of the COVID-19 pandemic, massive fiscal and monetary stimulus, subsequent multi-decade inflation surges, aggressive monetary policy tightening

by the Federal Reserve and Bangko Sentral ng Pilipinas (BSP), the rapid commercialization of artificial intelligence, and the institutional integration of digital assets.

Navigating this regime requires rigorous quantitative portfolio management. Modern Portfolio Theory (MPT), pioneered by Markowitz (1952) and extended by Sharpe (1964), provides the foundational framework for optimizing the trade-off between expected return and portfolio variance. However, empirical financial returns exhibit pronounced non-Gaussian characteristics—specifically heavy tails (leptokurtosis) and volatility clustering—demanding advanced downside risk analytics including Value at Risk (VaR), Expected Shortfall (CVaR), Monte Carlo simulation, and historical stress testing (Jorion, 2007; Artzner et al., 1999).

This report delivers a comprehensive portfolio consultation for Group 3 (Charlie). Section 2 details data acquisition, cleaning, and log return calculations. Section 3 presents exploratory financial analysis and asset distribution diagnostics. Section 4 evaluates equal-weighted buy-and-hold versus monthly rebalanced execution. Section 5 solves for the Efficient Frontier, GMV, and Max Sharpe portfolios. Section 6 conducts quantitative risk assessments, and Section 7 formulates strategic investment recommendations.

4 Data Ingestion and Preparation

Load the compiled master dataset containing historical daily close prices for Group 3's asset universe (TEL, MER, AEV, NVDA, META, BTC, SPY) and the risk-free rate benchmark over the sample period (January 2, 2020 – December 31, 2025).

```
# Define path to master dataset
data_path <- file.path("../", "data", "master_dataset.csv")

if (!file.exists(data_path)) {
  data_path <- file.path("data", "master_dataset.csv")
}

raw_data <- read_csv(data_path, show_col_types = FALSE)

# Filter dataset strictly to sample period Jan 2, 2020 to Dec 31, 2025
raw_data <- raw_data %>%
  mutate(Date = as.Date(Date)) %>%
  filter(Date >= as.Date("2020-01-02") & Date <= as.Date("2025-12-31")) %>%
  arrange(Date)

# Preview structure
glimpse(raw_data)
```

Rows: 1,508

Columns: 9

```
$ Date      <date> 2020-01-02, 2020-01-03, 2020-01-06, 2020-01-07, 2020-01-08,
  ↪ 2~
```

```

$ TEL      <dbl> 12.94961, 13.00212, 12.92992, 13.35654, 13.46156, 13.75691,
  ↪ 13~
$ MER      <dbl> 6.232742, 6.173474, 6.090038, 6.039285, 5.919388, 5.967083,
  ↪ 6.~
$ AEV      <dbl> 1.0305720, 1.0345115, 1.0339358, 1.0260920, 1.0169658,
  ↪ 1.02689~
$ NVDA     <dbl> 5.963804, 5.868347, 5.892957, 5.964300, 5.975487, 6.041113,
  ↪ 6.~
$ META     <dbl> 207.9538, 206.8535, 210.7493, 211.2053, 213.3465, 216.3997,
  ↪ 21~
$ BTC      <dbl> 6985.470, 7344.884, 7769.219, 8163.692, 8079.863, 7879.071,
  ↪ 81~
$ SPY      <dbl> 296.1253, 293.8830, 295.0042, 294.1747, 295.7425, 297.7478,
  ↪ 29~
$ RF_Rate  <dbl> 0.0525, 0.0525, 0.0525, 0.0525, 0.0525, 0.0525, 0.0525,
  ↪ 0.0525~

```

4.1 Daily Log Return Computation

Compute daily log returns for all portfolio assets and the benchmark:

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

```

prices_df <- raw_data

# Extract asset price columns
asset_cols <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC", "SPY")

# Convert matrix to xts time series object
prices_mat <- as.matrix(prices_df[, asset_cols])
prices_xts <- xts(prices_mat, order.by = prices_df[["Date"]])

returns_xts <- Return.calculate(prices_xts, method = "log")[-1, ]

# Separate portfolio assets from benchmark
portfolio_assets <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC")
benchmark_asset <- "SPY"

asset_returns_xts <- returns_xts[, portfolio_assets]
benchmark_returns_xts <- returns_xts[, benchmark_asset]

# No-trade days carry the last price mark, so the resumption-day return
# captures the full move across the gap. Wealth indexes use exp(cumsum(r))
# because these are log returns; cumprod(1 + r) would misstate growth.

```

```

prices_carried <- na.locf(prices_xts)
returns_xts_carried <- Return.calculate(prices_carried, method = "log")[-1,
  ↪ ]
asset_returns_carried <- returns_xts_carried[, portfolio_assets]

# EDA series: carried returns with no-trade days set back to NA, so
# resumption-day returns (full gap moves) enter the moments, correlations,
# and distribution figures while no-trade days stay missing.
returns_xts_eda <- returns_xts_carried
returns_xts_eda[is.na(prices_xts[-1, ])] <- NA
asset_returns_eda <- returns_xts_eda[, portfolio_assets]

# Risk-free rate (convert annual T-Bill rate to daily log-equivalent rate)
rf_annual <- mean(prices_df[["RF_Rate"]], na.rm = TRUE)
rf_daily <- log(1 + rf_annual) / 252

cat("Dataset span:", format(min(index(returns_xts))), "to",
  ↪ format(max(index(returns_xts))), "\n")

```

Dataset span: 2020-01-03 to 2025-12-31

```
cat("Total trading days:", nrow(returns_xts), "\n")
```

Total trading days: 1507

```
cat("Annualized Risk-Free Rate:", percent(rf_annual, accuracy = 0.01), "\n")
```

Annualized Risk-Free Rate: 5.25%

```
cat("Daily Log Risk-Free Rate:", percent(rf_daily, accuracy = 0.0001), "\n")
```

Daily Log Risk-Free Rate: 0.0203%

4.2 Data Validation and Summary Check

Verify data integrity, check for missing values, and print head records of returns.

```

missing_summary <- colSums(is.na(returns_xts))
notrade_summary <- colSums(is.na(returns_xts_eda))
cat("Missing return days per asset (raw gap-boundary convention):\n")

```

Missing return days per asset (raw gap-boundary convention):

```
print(missing_summary)
```

TEL	MER	AEV	NVDA	META	BTC	SPY
0	162	163	0	0	0	0

Table 1: First 6 Days of Asset Daily Log Returns (2020)

Date	TEL	MER	AEV	NVDA	META	BTC	SPY
2020-01-03	0.0040	-0.0096	0.0038	-0.0161	-0.0053	0.0502	-0.0076
2020-01-06	-0.0056	-0.0136	-0.0006	0.0042	0.0187	0.0562	0.0038
2020-01-07	0.0325	-0.0084	-0.0076	0.0120	0.0022	0.0495	-0.0028
2020-01-08	0.0078	-0.0201	-0.0089	0.0019	0.0101	-0.0103	0.0053
2020-01-09	0.0217	0.0080	0.0097	0.0109	0.0142	-0.0252	0.0068
2020-01-10	0.0029	0.0189	-0.0002	0.0053	-0.0011	0.0358	-0.0029

```
cat("No-trade days per asset (gap-aware convention):\n")
```

No-trade days per asset (gap-aware convention):

```
print(notrade_summary)
```

```
TEL  MER  AEV  NVDA  META  BTC  SPY
  0   86   87    0    0    0    0
```

```
if (any(missing_summary > 0)) {
  message("Missing return days present; verify no-trade day treatment in
  ↪ data acquisition")
}
```

```
# Format first 6 rows for table display
returns_head_df <- as.data.frame(head(returns_xts)) %>%
  rownames_to_column(var = "Date") %>%
  mutate(across(-Date, ~ sprintf("%.4f", .x)))
```

```
kbl(returns_head_df, booktabs = TRUE, align = "c") %>%
  kable_styling(latex_options = c("striped", "hold_position"),
  ↪ bootstrap_options = c("striped", "hover"))
```

5 Exploratory Financial Analysis

```
stats_mat <- table.Stats(asset_returns_eda)

# Convert to data frame for clean rendering
stats_df <- as.data.frame(stats_mat) %>%
  rownames_to_column(var = "Metric")

kbl(stats_df, booktabs = TRUE, digits = 4) %>%
```

Table 2: Descriptive Statistics for Portfolio Assets (Jan 2, 2020 – Dec 31, 2025)

Metric	TEL	MER	AEV	NVDA	META	BTC
Observations	1507.0000	1421.0000	1420.0000	1507.0000	1507.0000	1507.0000
NAs	0.0000	86.0000	87.0000	0.0000	0.0000	0.0000
Minimum	-0.2703	-0.1668	-0.1680	-0.2040	-0.3064	-0.4647
Quartile 1	-0.0098	-0.0106	-0.0141	-0.0157	-0.0117	-0.0161
Median	0.0000	-0.0001	0.0003	0.0032	0.0010	0.0010
Arithmetic Mean	0.0003	0.0003	-0.0005	0.0023	0.0008	0.0017
Geometric Mean	0.0001	0.0001	-0.0009	0.0017	0.0004	0.0009
Quartile 3	0.0101	0.0112	0.0127	0.0215	0.0142	0.0202
Maximum	0.1548	0.1175	0.1258	0.2181	0.2093	0.1915
SE Mean	0.0005	0.0005	0.0007	0.0009	0.0007	0.0010
LCL Mean (0.95)	-0.0007	-0.0007	-0.0019	0.0006	-0.0006	-0.0003
UCL Mean (0.95)	0.0013	0.0014	0.0008	0.0040	0.0022	0.0037
Variance	0.0004	0.0004	0.0006	0.0011	0.0008	0.0015
Stdev	0.0199	0.0204	0.0253	0.0333	0.0279	0.0393
Skewness	-1.7677	-0.2346	-0.2215	0.0439	-1.0740	-1.2216
Kurtosis	31.4634	5.7962	3.7706	4.0527	22.0895	16.2926

```
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
  ↪ bootstrap_options = c("striped", "hover"))
```

```
benchmark_stats_df <- as.data.frame(table.Stats(returns_xts_eda[, "SPY"]))
↪ %>%
  rownames_to_column(var = "Metric") %>%
  filter(Metric %in% c("Observations", "Arithmetic Mean", "Stdev",
    ↪ "Skewness", "Kurtosis", "Minimum", "Maximum")) %>%
  mutate(Metric = recode(Metric, "Arithmetic Mean" = "Mean"))

safe_kable(benchmark_stats_df, caption = "SPY Benchmark Descriptive
  ↪ Statistics (Daily Log Returns, Jan 2, 2020 -- Dec 31, 2025)")
```

Table 3: SPY Benchmark Descriptive Statistics (Daily Log Returns, Jan 2, 2020 – Dec 31, 2025)

Metric	SPY
Observations	1507.0000
Minimum	-0.1159
Mean	0.0005
Maximum	0.0999
Stdev	0.0131
Skewness	-0.5584

5.1 Quantitative Design System (QVRS Theme & Colors)

```
# QVRS Economist Color Tokens
qvrs_colors <- c(
  "TEL" = "#1F2E7A", # Chicago 30
  "MER" = "#2E45B8", # Chicago 45
  "AEV" = "#475ED1", # Chicago 55
  "NVDA" = "#1DC9A4", # Hong Kong 45
  "META" = "#F97A1F", # Singapore 55
  "BTC" = "#E3120B", # Economist Red
  "SPY" = "#595959" # London 35
)

# Custom Economist/QVRS ggplot2 theme
theme_quant <- function() {
  theme_minimal() +
    theme(
      text = element_text(color = "#0D0D0D"),
      plot.title = element_text(face = "bold", size = 12, margin = margin(b
        → = 4)),
      plot.subtitle = element_text(size = 9.5, color = "#333333", margin =
        → margin(b = 8)),
      plot.caption = element_text(size = 8, color = "#595959", hjust = 0,
        → margin = margin(t = 6)),
      axis.title = element_text(size = 9, face = "bold", color = "#1A1A1A"),
      axis.text = element_text(size = 8, color = "#333333"),
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "#F2F2F2", linewidth = 0.3),
      legend.position = "bottom",
      legend.title = element_blank(),
      axis.ticks = element_blank(),
      strip.background = element_rect(fill = "#EBEDFA", color = NA),
      strip.text = element_text(face = "bold", size = 9, color = "#141F52")
    )
}
```

5.2 Historical Price & Return Trajectories

```
# Compute normalized price index (Base = 100 at start of sample period)
prices_norm_df <- as.data.frame(prices_carried) %>%
  rownames_to_column(var = "Date") %>%
```

```

mutate(Date = as.Date(Date)) %>%
pivot_longer(-Date, names_to = "Asset", values_to = "Price") %>%
group_by(Asset) %>%
mutate(Normalized_Price = (Price / first(Price)) * 100) %>%
ungroup() %>%
mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(prices_norm_df, aes(x = Date, y = Normalized_Price, color = Asset)) +
  geom_line(linewidth = 0.75, alpha = 0.9) +
  scale_color_manual(values = qvrs_colors) +
  scale_y_log10(labels = label_comma(prefix = "$")) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Historical Normalized Asset Price Trajectories (Log Scale)",
    subtitle = "Base = 100 on January 2, 2020 through December 31, 2025",
    x = NULL,
    y = "Normalized Index (Jan 2020 = 100)"
  ) +
  theme_quant()

```

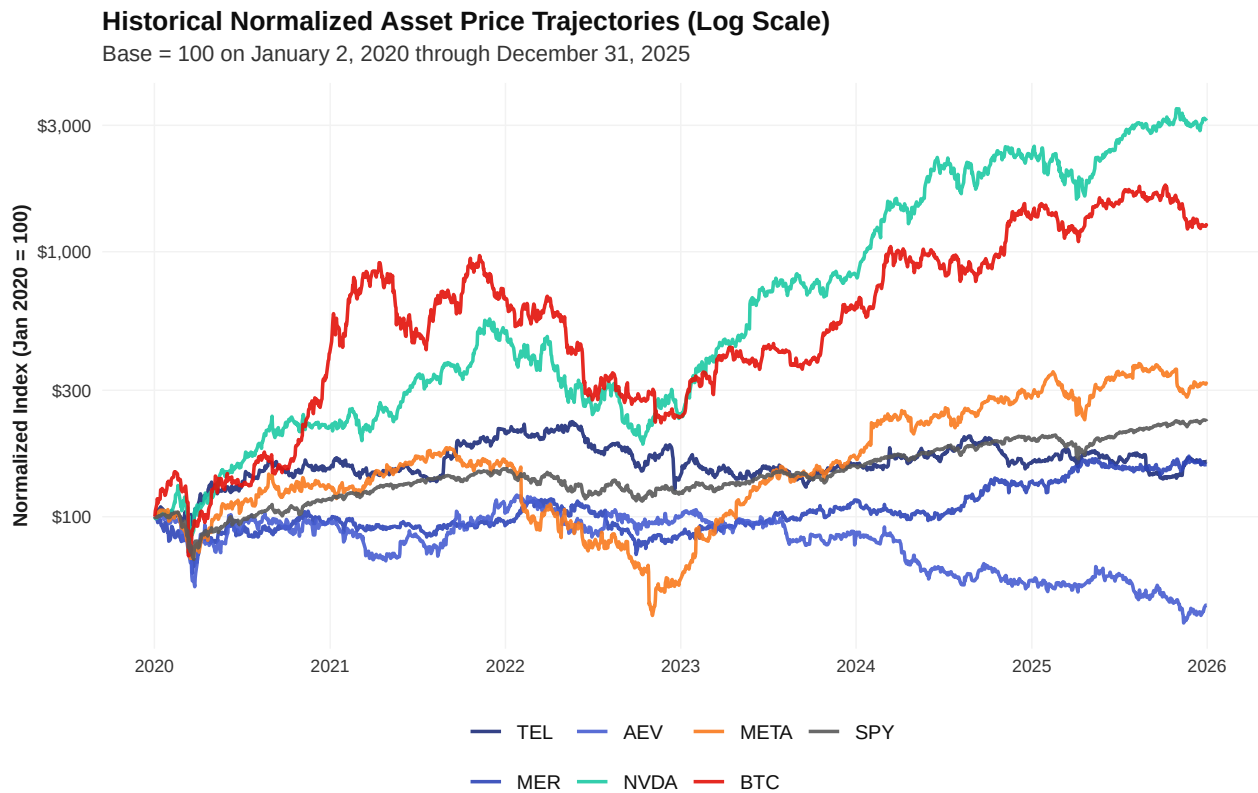


Figure 1: Historical Normalized Asset Price Trajectories (Base = 100 on Jan 2, 2020)

```

# Reshape returns for tidy plotting
returns_tidy <- as.data.frame(returns_xts_eda) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Return") %>%
  mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(returns_tidy, aes(x = Date, y = Return, color = Asset)) +
  geom_line(linewidth = 0.35, alpha = 0.85) +
  facet_wrap(~ Asset, ncol = 2, scales = "free_y") +
  scale_color_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  scale_x_date(date_labels = "%Y", date_breaks = "2 years") +
  labs(
    title = "Daily Log Return Dynamics and Volatility Clustering",
    subtitle = "Sample Period: January 2, 2020 through December 31, 2025",
    x = NULL,
    y = "Daily Log Return"
  ) +
  theme_quant() +
  theme(legend.position = "none")

```

Daily Log Return Dynamics and Volatility Clustering

Sample Period: January 2, 2020 through December 31, 2025

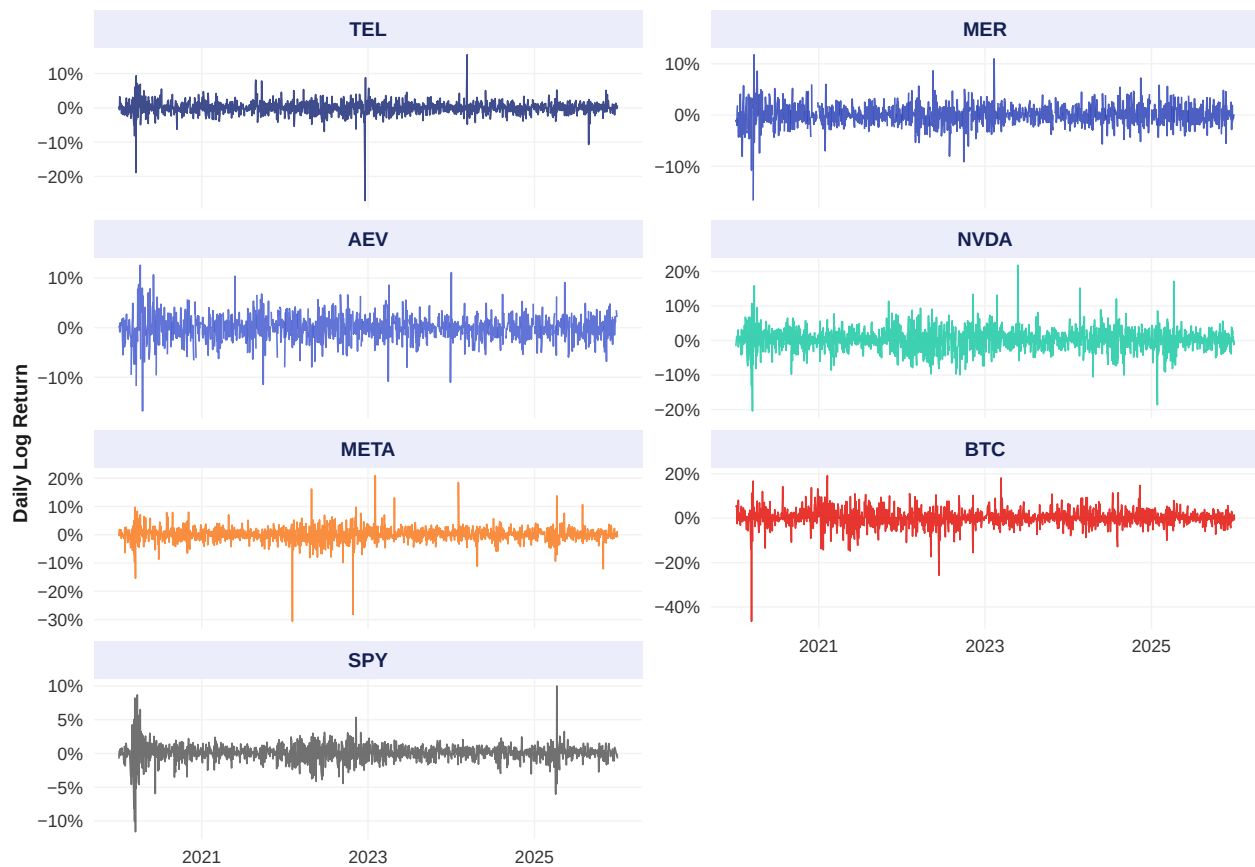


Figure 2: Daily Log Return Trajectories Across Portfolio Universe (Jan 2020 – Dec 2025)

5.3 Distributional Shape: Histograms and Empirical Density vs Normal Reference

```
# Compute normal distribution parameters per asset for reference curves
norm_params <- returns_tidy %>%
  group_by(Asset) %>%
  summarise(
    mean = mean(Return, na.rm = TRUE),
    sd = sd(Return, na.rm = TRUE),
    .groups = "drop"
  )

# Grid of x values for theoretical normal density overlay
grid_df <- norm_params %>%
  group_by(Asset) %>%
  reframe(
    x = seq(mean - 4 * sd, mean + 4 * sd, length.out = 200),
```

```

    y = dnorm(x, mean = mean, sd = sd)
  )

ggplot(returns_tidy, aes(x = Return)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 60,
    fill = "#D6DBF5",
    color = "#1F2E7A",
    linewidth = 0.2,
    alpha = 0.7
  ) +
  geom_density(aes(color = "Empirical Density"), linewidth = 0.8) +
  geom_line(
    data = grid_df,
    aes(x = x, y = y, color = "Gaussian Normal"),
    linetype = "dashed",
    linewidth = 0.75
  ) +
  facet_wrap(~ Asset, scales = "free", ncol = 3) +
  scale_color_manual(
    values = c("Empirical Density" = "#141F52", "Gaussian Normal" =
      ↪ "#E3120B")
  ) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Empirical Distribution Shape: Heavy Tails and Non-Normality",
    subtitle = "Comparison of empirical kernel density against theoretical
      ↪ normal distribution",
    x = "Daily Log Return",
    y = "Density"
  ) +
  theme_quant()

```

Empirical Distribution Shape: Heavy Tails and Non-Normality

Comparison of empirical kernel density against theoretical normal distribution

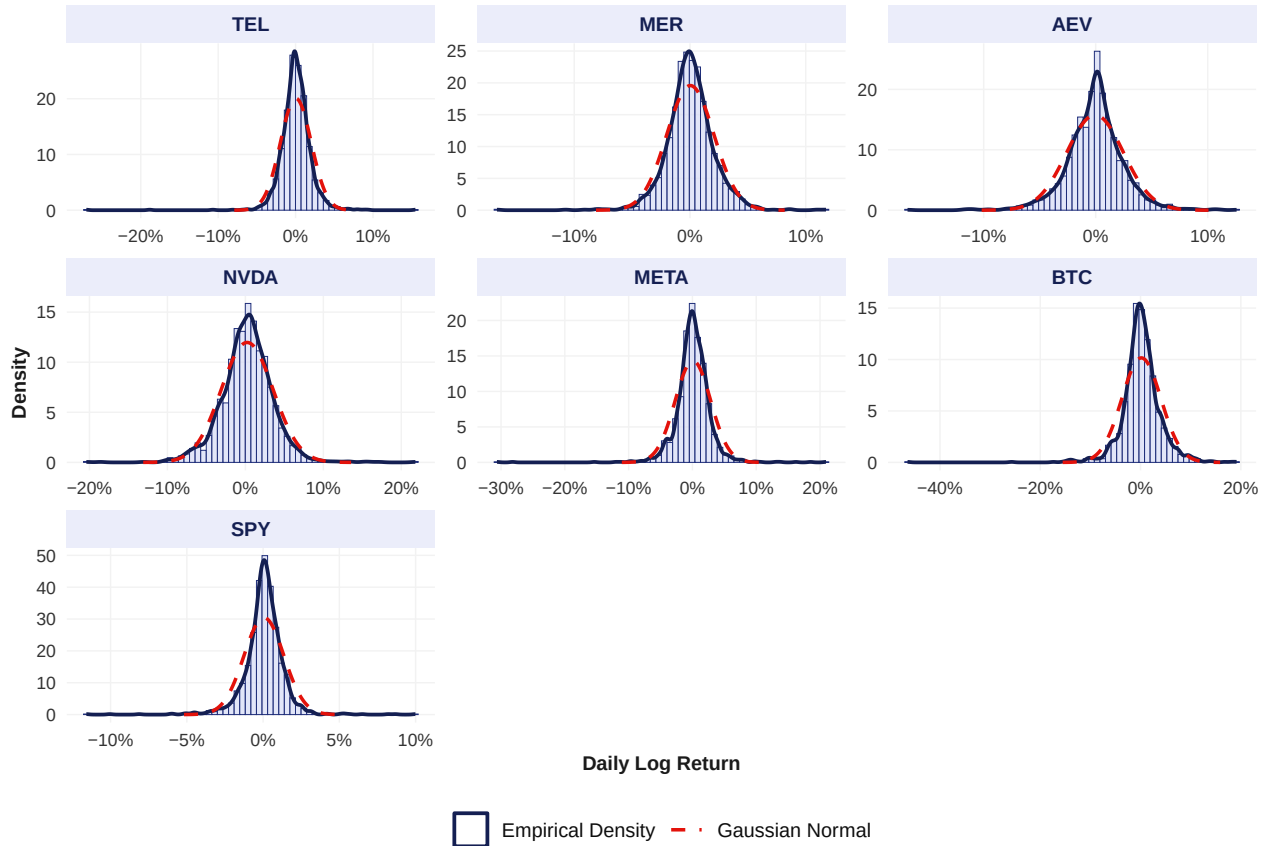


Figure 3: Empirical Return Distributions vs Gaussian Reference Curves

5.4 Quantile-Quantile (Q-Q) Tail Diagnostics

```
ggplot(returns_tidy, aes(sample = Return)) +
  stat_qq_line(color = "#E3120B", linewidth = 0.7, linetype = "dashed") +
  stat_qq(color = "#1F2E7A", alpha = 0.5, size = 0.8) +
  facet_wrap(~ Asset, scales = "free", ncol = 3) +
  labs(
    title = "Quantile-Quantile Tail Departure from Normality",
    subtitle = "S-shaped curves confirm pronounced fat-tail behavior across
      ↪ asset returns",
    x = "Theoretical Gaussian Quantiles",
    y = "Sample Quantiles"
  ) +
  theme_quant()
```

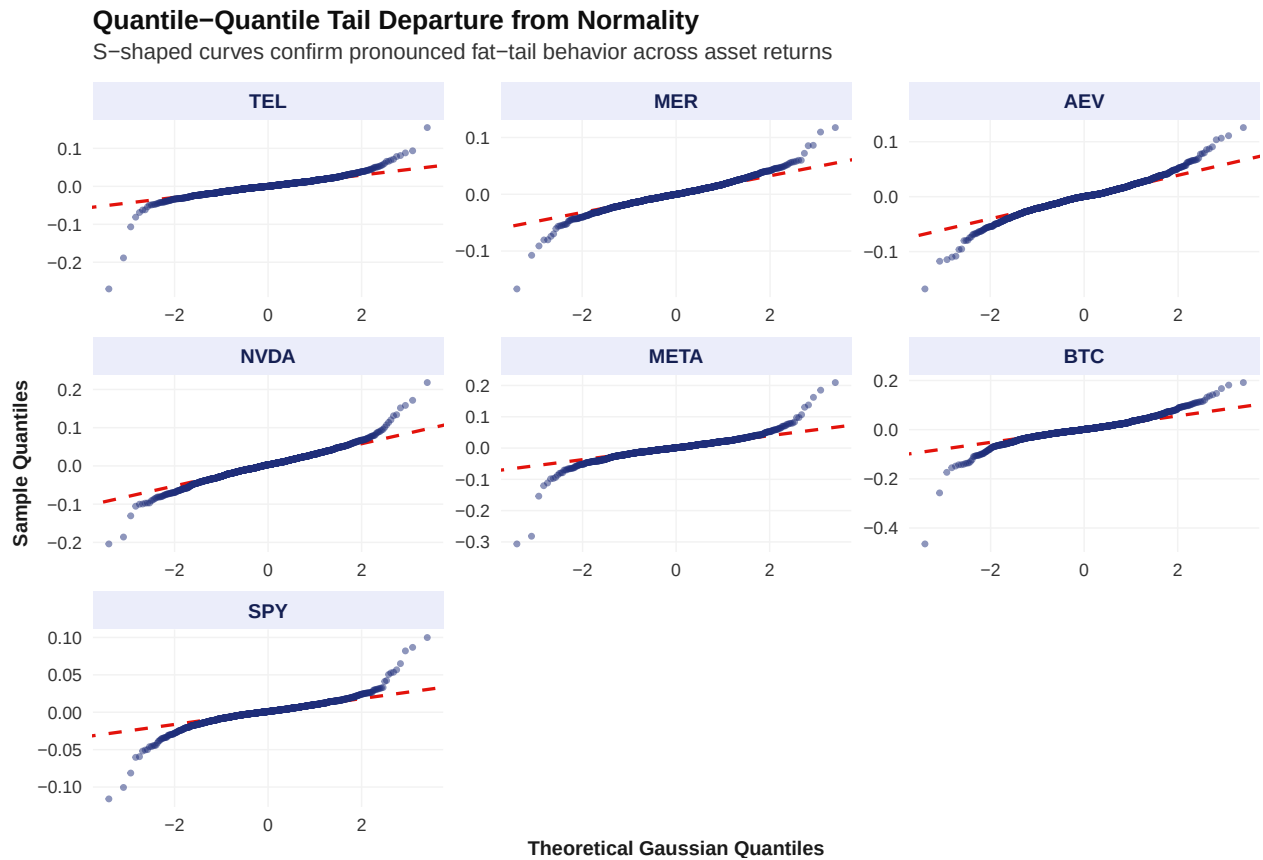


Figure 4: Normal Q-Q Diagnostics Demonstrating Heavy Tail Leptokurtosis

5.5 Dispersion of Daily Returns and Outlier Diagnostics

```
ggplot(returns_tidy, aes(x = Asset, y = Return, fill = Asset)) +
  geom_boxplot(outlier.color = "#E3120B", outlier.size = 1, outlier.alpha =
    ↪ 0.5, linewidth = 0.4) +
  scale_fill_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Cross-Asset Return Dispersion and Extreme Tail Outliers",
    subtitle = "Comparison of median, interquartile range (IQR), and extreme
    ↪ shocks",
    x = "Asset",
    y = "Daily Log Return"
  ) +
  theme_quant() +
  theme(legend.position = "none")
```

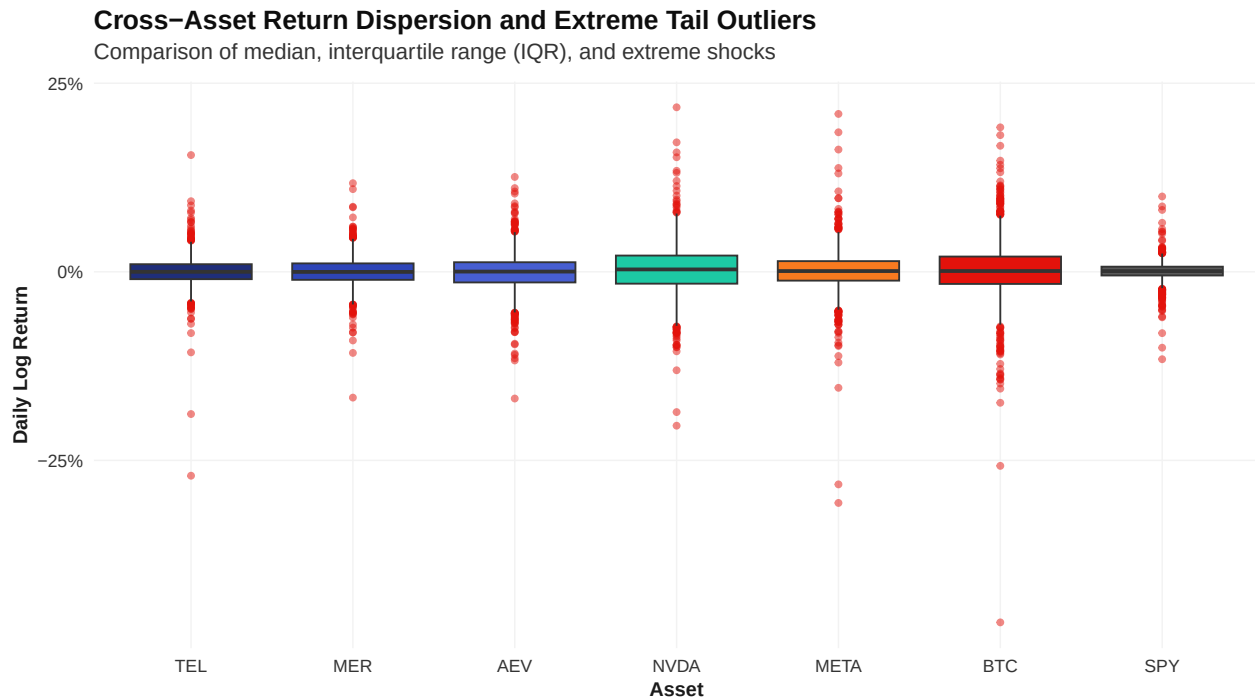


Figure 5: Return Dispersion and Extreme Outlier Boxplots

5.6 Cumulative Performance Comparison (Growth of \$1.00)

```
cum_returns_df <- as.data.frame(exp(cumsum(returns_xts_carried))) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Wealth") %>%
  mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(cum_returns_df, aes(x = Date, y = Wealth, color = Asset)) +
  geom_line(linewidth = 0.75) +
  scale_color_manual(values = qvrs_colors) +
  scale_y_log10(labels = label_dollar(prefix = "USD ")) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Cumulative Wealth Index Growth Trajectories (Log Scale)",
    subtitle = "Hypothetical USD 1.00 investment performance from January
  ↪ 2020 to December 2025",
    x = NULL,
    y = "Wealth Index (USD)"
  ) +
  theme_quant()
```

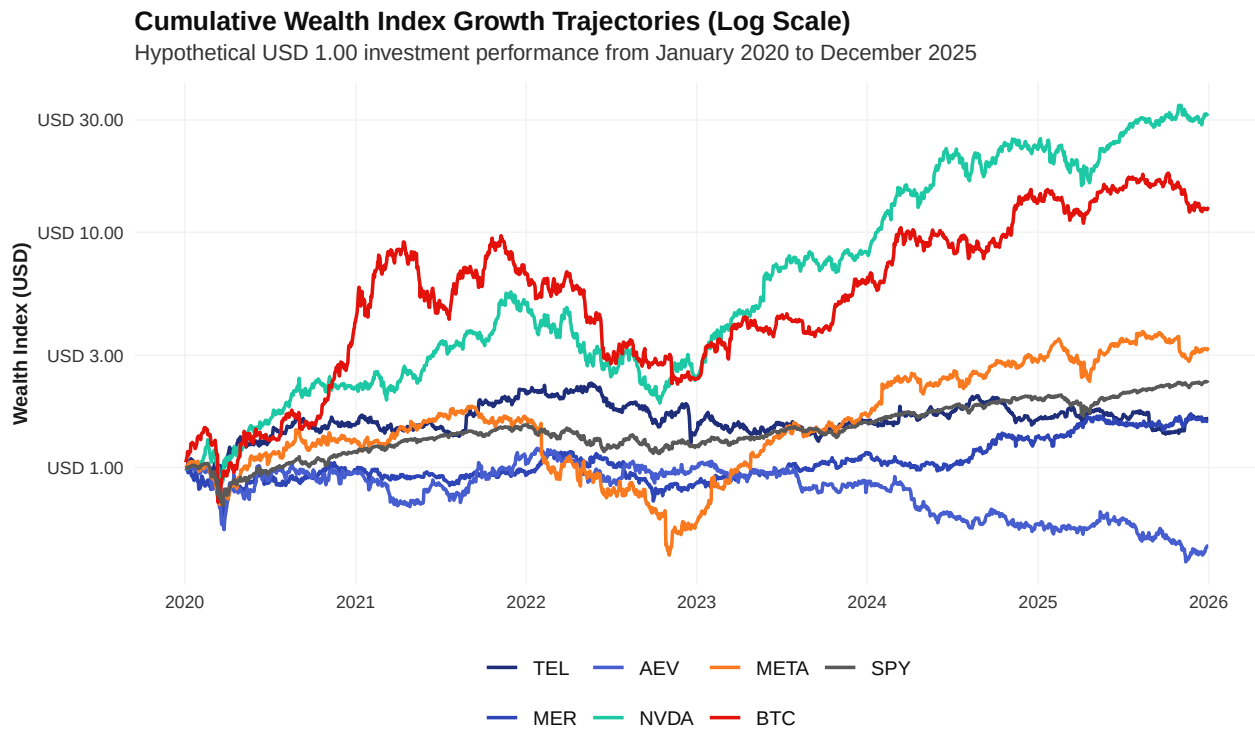


Figure 6: Cumulative Growth Trajectories of USD 1.00 Investment (Jan 2020 – Dec 2025)

5.7 Correlation Structure

```
cor_matrix <- cor(returns_xts_eda, use = "pairwise.complete.obs")

# Render correlation heatmap
corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  tl.col = "black",
  number.cex = 0.8,
  mar = c(0, 0, 1, 0)
)
```

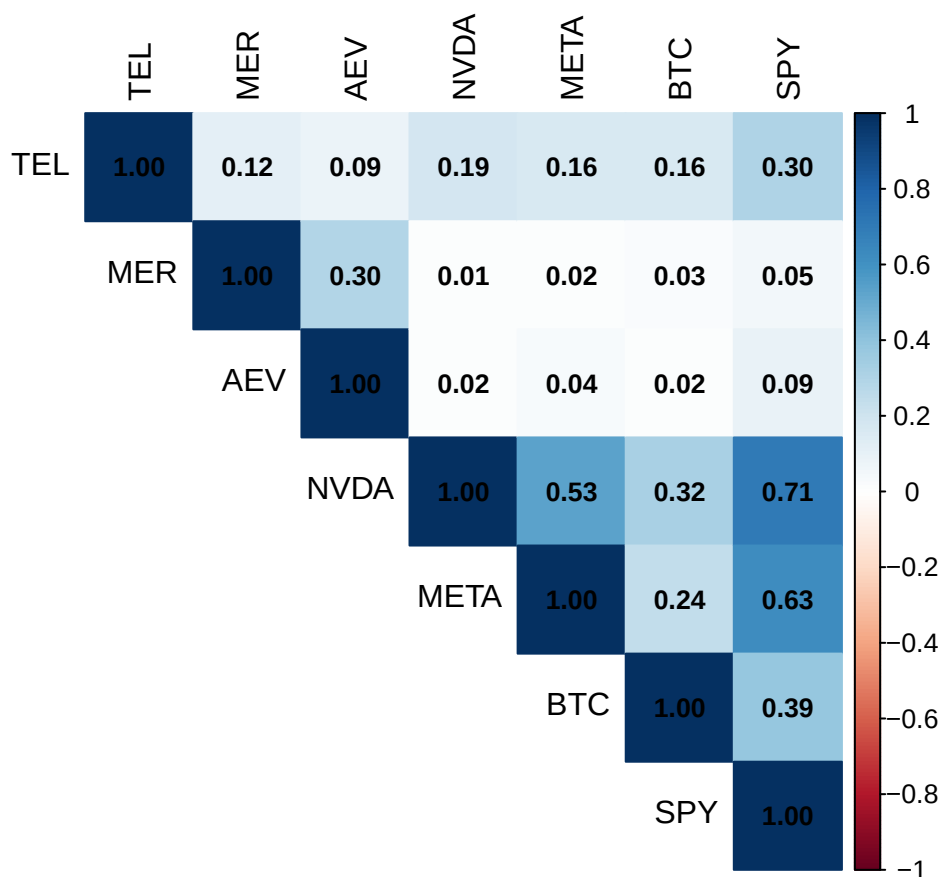


Figure 7: Asset Return Correlation Heatmap (Group 3 Universe)

6 Portfolio Construction (Equal-Weight vs Monthly Rebalanced)

This section evaluates the performance of equal-weighted asset allocations under two distinct execution regimes: a static Buy-and-Hold strategy and a disciplined Monthly Rebalanced strategy. The 6-asset portfolio universe (TEL, MER, AEV, NVDA, META, BTC) starts with equal baseline weights ($w_i = 1/6 \approx 16.67\%$) on January 2, 2020 and is benchmarked against the SPDR S&P 500 ETF Trust (SPY).

```
# Simple (arithmetic) returns: a weighted sum of log returns is not the
# portfolio return, so Return.portfolio and all portfolio metrics below
# operate on simple returns converted from the carried log series.
ret_smp <- exp(returns_xts_carried) - 1
asset_returns_smp <- ret_smp[, portfolio_assets]

# Equal weight allocation vector across the 6 portfolio assets
w_eq <- rep(1 / 6, 6)
names(w_eq) <- colnames(asset_returns_smp)
```

```

# Strategy 1: Buy-and-Hold Equal-Weighted Portfolio
port_bnh <- Return.portfolio(
  R = asset_returns_smp,
  weights = w_eq,
  verbose = TRUE
)
r_bnh <- port_bnh$returns
weights_bnh <- port_bnh$EOP.Weight

# Strategy 2: Monthly Rebalanced Equal-Weighted Portfolio
port_reb <- Return.portfolio(
  R = asset_returns_smp,
  weights = w_eq,
  rebalance_on = "months",
  verbose = TRUE
)
r_reb <- port_reb$returns
weights_reb <- port_reb$EOP.Weight

# Risk-free rate metrics (5.25% per annum)
rf_annual <- 0.0525
rf_daily <- log(1 + rf_annual) / 252

# Performance evaluation helper function
eval_portfolio <- function(r_series, name) {
  cum_ret <- as.numeric(tail(cumprod(1 + r_series), 1)) - 1
  n_obs <- length(r_series)
  ann_ret <- (1 + cum_ret)^(252 / n_obs) - 1
  ann_sd <- sd(as.numeric(r_series)) * sqrt(252)
  sharpe <- (ann_ret - rf_annual) / ann_sd
  max_dd <- as.numeric(maxDrawdown(r_series))

  data.frame(
    Strategy = name,
    `Cumulative Return` = sprintf("%.2f%%", cum_ret * 100),
    `CAGR (Annualized)` = sprintf("%.2f%%", ann_ret * 100),
    `Annual Volatility` = sprintf("%.2f%%", ann_sd * 100),
    `Sharpe Ratio (Rf=5.25%)` = sprintf("%.4f", sharpe),
    `Max Drawdown` = sprintf("%.2f%%", max_dd * 100),
    check.names = FALSE
  )
}

perf_summary_df <- rbind(
  eval_portfolio(r_bnh, "Equal-Weight (Buy & Hold)"),

```

```
eval_portfolio(r_reb, "Equal-Weight (Monthly Rebalanced)",
eval_portfolio(ret_smp[, "SPY"], "SPDR S&P 500 ETF (SPY Benchmark)")
)
```

6.1 Comparative Performance Evaluation

```
safe_kable(perf_summary_df, caption = "Portfolio Performance Summary
↪ Comparison (Jan 2, 2020 -- Dec 31, 2025)")
```

Table 4: Portfolio Performance Summary Comparison (Jan 2, 2020 – Dec 31, 2025)

Strategy	Cumulative Return	CAGR (Annualized)	Annual Volatility	Sharpe Ratio (Rf=5.25%)	Max
Equal-Weight (Buy & Hold)	743.04%	42.83%	35.12%	1.0700	58.21
Equal-Weight (Monthly Rebalanced)	384.85%	30.21%	24.18%	1.0325	43.20
SPDR S&P 500 ETF (SPY Benchmark)	129.06%	14.87%	20.75%	0.4635	33.72

6.2 Calendar-Year Return Decomposition

```
yearly_ret <- function(r_series, yr) {
  rr <- r_series[format(index(r_series), "%Y") == yr]
  as.numeric(tail(cumprod(1 + rr), 1)) - 1
}

cyr_strategy_df <- do.call(rbind, lapply(2020:2025, function(yr) {
  data.frame(
    Year = yr,
    `Buy & Hold` = sprintf("%.2f%%", yearly_ret(r_bnh, yr) * 100),
    `Monthly Rebalanced` = sprintf("%.2f%%", yearly_ret(r_reb, yr) * 100),
    `SPY Benchmark` = sprintf("%.2f%%", yearly_ret(ret_smp[, "SPY"], yr) *
↪ 100),
    check.names = FALSE
  )
}))

safe_kable(cyr_strategy_df, caption = "Calendar-Year Strategy Returns
↪ (Simple Returns, 2020--2025)")
```

Table 5: Calendar-Year Strategy Returns (Simple Returns, 2020–2025)

Year	Buy & Hold	Monthly Rebalanced	SPY Benchmark
2020	84.72%	70.37%	17.24%
2021	54.93%	46.80%	28.73%
2022	-49.57%	-36.77%	-18.18%

2023	125.47%	85.09%	26.18%
2024	116.05%	50.88%	24.89%
2025	19.92%	9.78%	17.72%

```
cyr_asset_df <- do.call(rbind, lapply(2020:2025, function(yr) {
  data.frame(
    Year = yr,
    TEL = sprintf("%.2f%%", yearly_ret(ret_smp[, "TEL"], yr) * 100),
    MER = sprintf("%.2f%%", yearly_ret(ret_smp[, "MER"], yr) * 100),
    AEV = sprintf("%.2f%%", yearly_ret(ret_smp[, "AEV"], yr) * 100),
    NVDA = sprintf("%.2f%%", yearly_ret(ret_smp[, "NVDA"], yr) * 100),
    META = sprintf("%.2f%%", yearly_ret(ret_smp[, "META"], yr) * 100),
    BTC = sprintf("%.2f%%", yearly_ret(ret_smp[, "BTC"], yr) * 100),
    check.names = FALSE
  )
}))

safe_kable(cyr_asset_df, caption = "Calendar-Year Asset Returns (Simple
↪ Returns, 2020--2025)")
```

Table 6: Calendar-Year Asset Returns (Simple Returns, 2020–2025)

Year	TEL	MER	AEV	NVDA	META	BTC
2020	51.43%	-2.20%	-4.29%	118.02%	30.21%	315.17%
2021	36.56%	-5.47%	7.75%	125.48%	23.13%	59.67%
2022	-31.70%	-7.63%	-3.29%	-50.26%	-64.22%	-64.15%
2023	11.92%	35.53%	-21.55%	239.02%	194.13%	153.57%
2024	0.81%	16.71%	-26.51%	171.25%	66.05%	121.93%
2025	1.48%	16.38%	-19.34%	38.92%	13.09%	-6.34%

6.3 Cumulative Portfolio Growth Comparison

```
# Merge portfolio return series (all simple returns)
port_returns_combined <- cbind(r_bnh, r_reb, ret_smp[, "SPY"])
colnames(port_returns_combined) <- c("Buy_Hold", "Monthly_Rebalanced",
↪ "SPY_Benchmark")

port_growth_df <- as.data.frame(cumprod(1 + port_returns_combined)) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Strategy", values_to = "Wealth") %>%
  mutate(
    Strategy = case_when(
      Strategy == "Buy_Hold" ~ "Equal-Weight (Buy & Hold)",
```

```

    Strategy == "Monthly_Rebalanced" ~ "Equal-Weight (Monthly
↪ Rebalanced)",
    Strategy == "SPY_Benchmark" ~ "SPDR S&P 500 ETF (SPY)"
  ),
  Strategy = factor(
    Strategy,
    levels = c("Equal-Weight (Buy & Hold)", "Equal-Weight (Monthly
↪ Rebalanced)", "SPDR S&P 500 ETF (SPY)")
  )
)

strat_colors <- c(
  "Equal-Weight (Buy & Hold)" = "#E3120B",          # Economist Red
  "Equal-Weight (Monthly Rebalanced)" = "#1F2E7A", # Chicago 30
  "SPDR S&P 500 ETF (SPY)" = "#595959"             # London 35
)

ggplot(port_growth_df, aes(x = Date, y = Wealth, color = Strategy, linetype
↪ = Strategy)) +
  geom_line(linewidth = 0.85) +
  scale_color_manual(values = strat_colors) +
  scale_linetype_manual(values = c("solid", "solid", "dashed")) +
  scale_y_log10(labels = label_dollar(prefix = "USD ")) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Cumulative Wealth Growth Comparison (Log Scale)",
    subtitle = "Performance of USD 1.00 allocation from January 2020 through
↪ December 2025",
    x = NULL,
    y = "Wealth Index (USD)"
  ) +
  theme_quant()

```

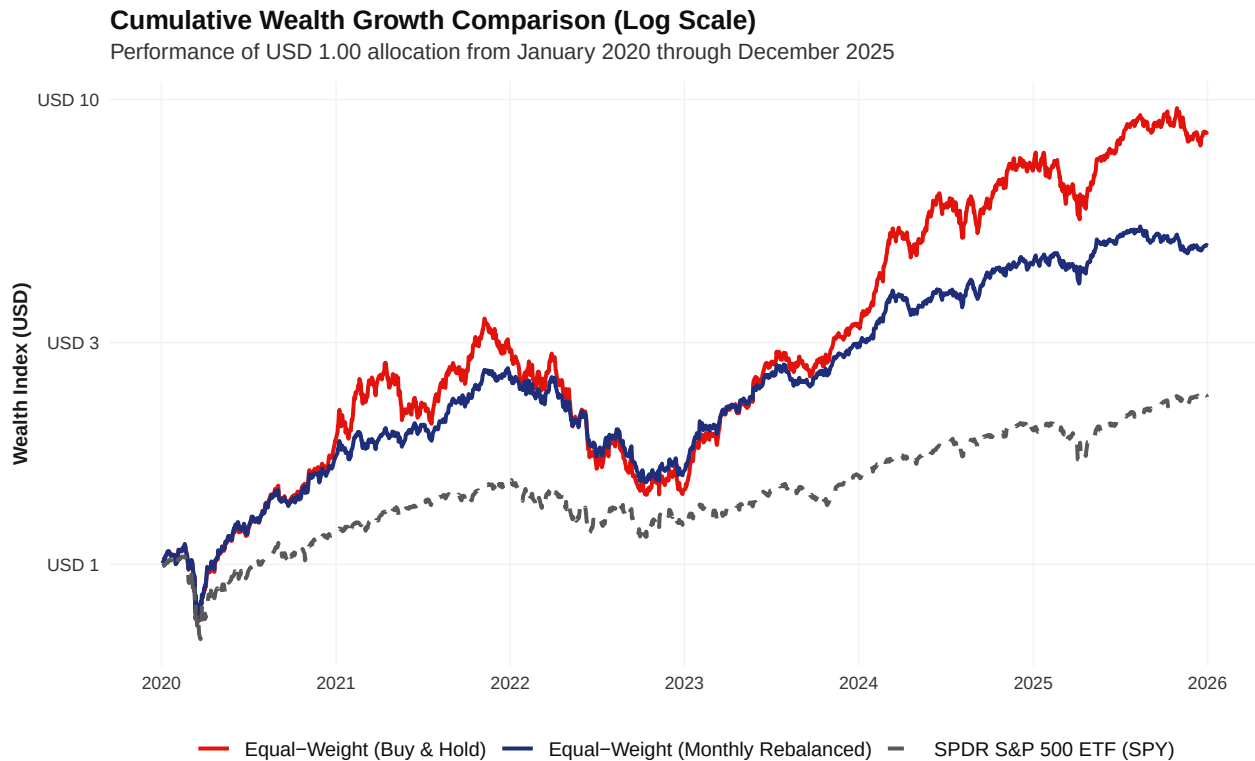


Figure 8: Cumulative Growth Trajectories of USD 1.00 Investment Across Strategies

6.4 Portfolio Asset Weight Drift Analysis

```
# Reshape end-of-period weights for Buy & Hold strategy
weights_bnh_df <- as.data.frame(weights_bnh) %>%
  rownames_to_column(var = "Date") %>%
  mutate(Date = as.Date(Date)) %>%
  pivot_longer(-Date, names_to = "Asset", values_to = "Weight") %>%
  mutate(Asset = factor(Asset, levels = names(qvrs_colors)))

ggplot(weights_bnh_df, aes(x = Date, y = Weight, fill = Asset)) +
  geom_area(alpha = 0.85, color = "#FFFFFF", linewidth = 0.1) +
  scale_fill_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  scale_x_date(date_labels = "%Y", date_breaks = "1 year") +
  labs(
    title = "Asset Allocation Drift in Buy-and-Hold Execution",
    subtitle = "Evolution of asset weights driven by differential market
    ↪ performance (Jan 2020 -- Dec 2025)",
    x = NULL,
    y = "Portfolio Weight Share"
  ) +
  theme_quant()
```

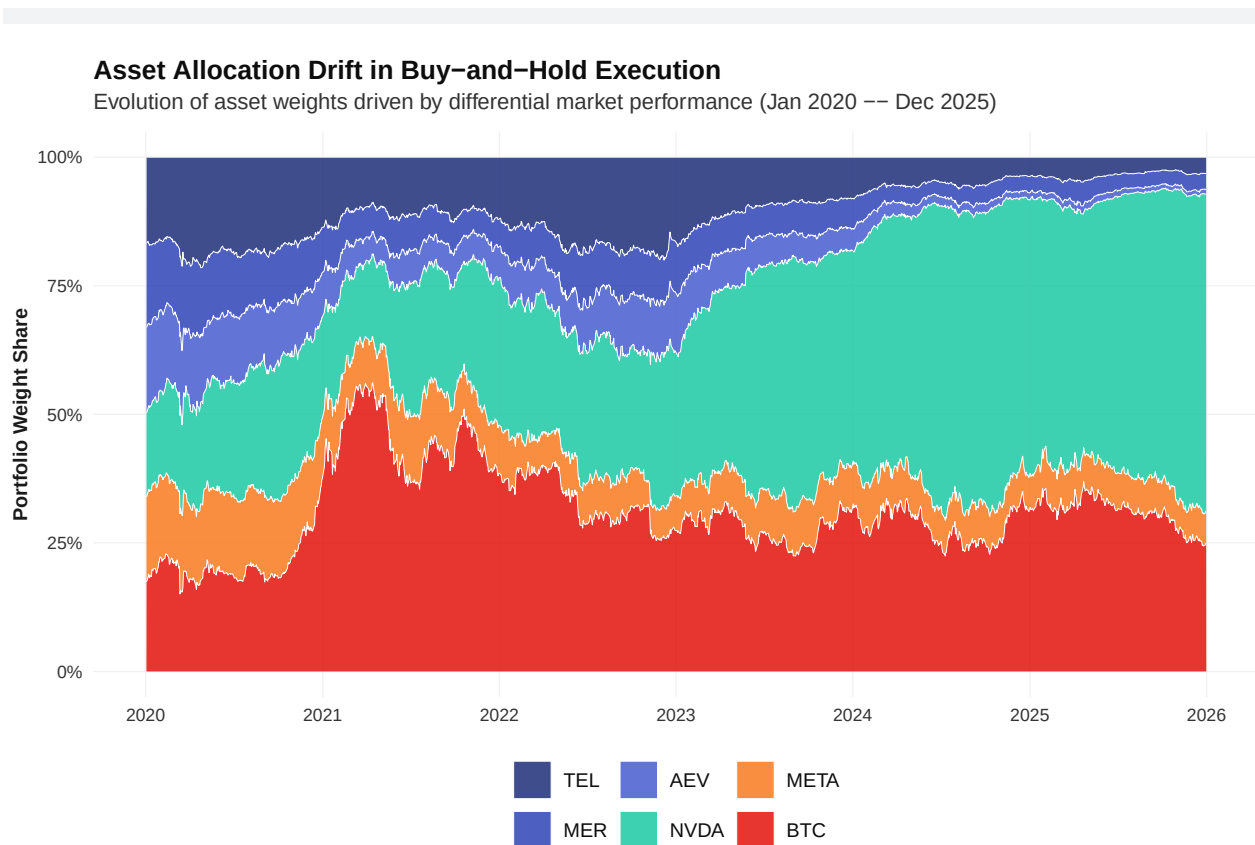


Figure 9: Asset Weight Drift Over Time in the Un-Rebalanced Buy-and-Hold Portfolio

7 Portfolio Optimization (Efficient Frontier, GMV, Max Sharpe)

```
mu_ann    <- colMeans(asset_returns_smp, na.rm = TRUE) * 252
Sigma_ann <- cov(asset_returns_smp, use = "pairwise.complete.obs") * 252

n_assets  <- length(mu_ann)
asset_names <- names(mu_ann)

# Equal-weight starting point for the solver
w0 <- rep(1 / n_assets, n_assets)

eval_f_var <- function(w) {
  as.numeric(t(w) %*% Sigma_ann %*% w)
}

eval_grad_f_var <- function(w) {
  nl.grad(w, eval_f_var)
```

```

}

eval_f_sharpe <- function(w) {
  port_ret <- sum(w * mu_ann)
  port_sd <- sqrt(as.numeric(t(w) %*% Sigma_ann %*% w))
  -(port_ret - rf_annual) / port_sd # negative since nloptr always
  ↪ minimizes
}

eval_grad_f_sharpe <- function(w) {
  nl.grad(w, eval_f_sharpe)
}

eval_g_eq_sum1 <- function(w) {
  sum(w) - 1
}

eval_jac_g_eq_sum1 <- function(w) {
  nl.jacobian(w, eval_g_eq_sum1)
}

# Solver settings: SLSQP as guided, since it's the R equivalent of Excel's
↪ GRG Nonlinear
opt_settings <- list(
  "algorithm" = "NLOPT_LD_SLSQP",
  "xtol_rel" = 1.0e-8,
  "maxeval" = 1000
)

gmw_solution <- nloptr(
  x0 = w0,
  eval_f = eval_f_var,
  eval_grad_f = eval_grad_f_var,
  lb = rep(0, n_assets), # long-only, no shorting
  ub = rep(1, n_assets), # no single asset above 100%
  eval_g_eq = eval_g_eq_sum1,
  eval_jac_g_eq = eval_jac_g_eq_sum1,
  opts = opt_settings
)

w_gmv <- gmw_solution$solution
names(w_gmv) <- asset_names

gmw_ret <- sum(w_gmv * mu_ann)
gmw_sd <- sqrt(as.numeric(t(w_gmv) %*% Sigma_ann %*% w_gmv))

```

```

sharpe_solution <- nloptr(
  x0          = w0,
  eval_f      = eval_f_sharpe,
  eval_grad_f = eval_grad_f_sharpe,
  lb          = rep(0, n_assets),
  ub          = rep(1, n_assets),
  eval_g_eq   = eval_g_eq_sum1,
  eval_jac_g_eq = eval_jac_g_eq_sum1,
  opts        = opt_settings
)

w_sharpe <- sharpe_solution$solution
names(w_sharpe) <- asset_names

sharpe_ret <- sum(w_sharpe * mu_ann)
sharpe_sd  <- sqrt(as.numeric(t(w_sharpe) %*% Sigma_ann %*% w_sharpe))
sharpe_val <- (sharpe_ret - rf_annual) / sharpe_sd

target_returns <- seq(min(mu_ann), max(mu_ann), length.out = 50)
frontier_list  <- vector("list", length(target_returns))

D <- 2 * Sigma_ann + diag(1e-6, n_assets)
d <- rep(0, n_assets)

for (i in seq_along(target_returns)) {
  target <- target_returns[i]

  Amat <- cbind(
    rep(1, n_assets), # sum(w) = 1
    mu_ann,           # sum(w * mu_ann) = target
    diag(n_assets),   # w >= 0
    -diag(n_assets)   # -w >= -1
  )
  bvec <- c(1, target, rep(0, n_assets), rep(-1, n_assets))

  sol <- tryCatch(
    quadprog::solve.QP(Dmat = D, dvec = d, Amat = Amat, bvec = bvec, meq =
→ 2),
    error = function(e) NULL
  )

  if (!is.null(sol)) {
    w_i <- sol$solution
    frontier_list[[i]] <- data.frame(

```

```

    Target_Return = target,
    Risk    = sqrt(as.numeric(t(w_i) %*% Sigma_ann %*% w_i)),
    Return = sum(w_i * mu_ann)
  )
}
}

efficient_frontier_df <- bind_rows(frontier_list)

efficient_only_df <- efficient_frontier_df[efficient_frontier_df$Return >=
  ↪  gmv_ret, ]

ggplot(efficient_only_df, aes(x = Risk, y = Return)) +
  geom_path(color = "#1F2E7A", linewidth = 0.9) +
  geom_point(
    data = data.frame(Risk = gmv_sd, Return = gmv_ret),
    aes(x = Risk, y = Return), color = "#E3120B", size = 3
  ) +
  geom_point(
    data = data.frame(Risk = sharpe_sd, Return = sharpe_ret),
    aes(x = Risk, y = Return), color = "#1DC9A4", size = 3
  ) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Efficient Frontier: Risk-Return Tradeoff",
    subtitle = "Red point = Global Minimum Variance | Teal point = Maximum
  ↪  Sharpe Ratio",
    x = "Annualized Volatility (Risk)",
    y = "Annualized Expected Return"
  ) +
  theme_quant()

```

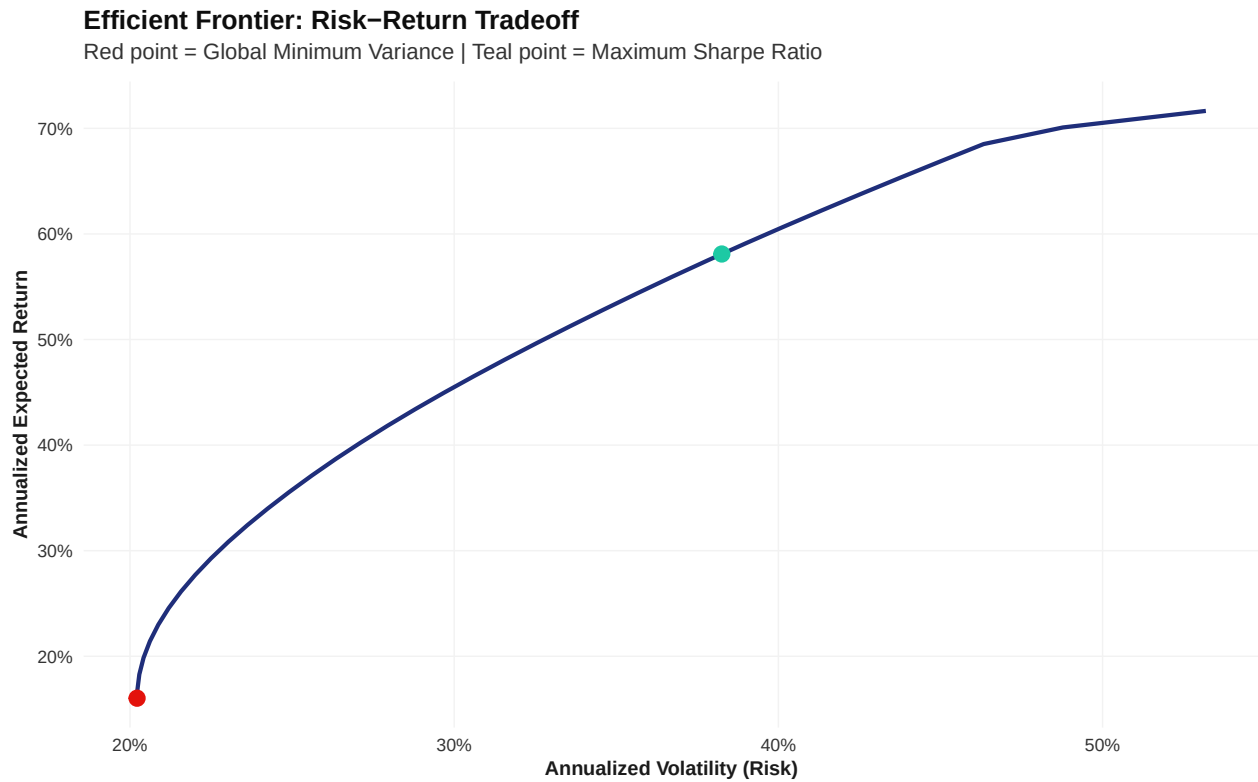


Figure 10: Efficient Frontier with Global Minimum Variance and Maximum Sharpe Portfolios

```
optimal_weights_df <- data.frame(
  Asset      = asset_names,
  GMV_Weight = sprintf("%.2f%%", w_gmv * 100),
  MaxSharpe_Weight = sprintf("%.2f%%", w_sharpe * 100)
)

safe_kable(optimal_weights_df, caption = "Optimal Portfolio Weights: GMV vs
↪ Maximum Sharpe")
```

Table 7: Optimal Portfolio Weights: GMV vs Maximum Sharpe

Asset	GMV_Weight	MaxSharpe_Weight
TEL	31.01%	0.00%
MER	31.23%	18.42%
AEV	16.85%	0.00%
NVDA	3.15%	55.44%
META	12.86%	0.00%
BTC	4.90%	26.14%

```
weights_long_df <- data.frame(
  Asset = rep(asset_names, 2),
```

```

Portfolio = rep(c("GMV", "Max Sharpe"), each = n_assets),
Weight     = c(w_gmv, w_sharpe)
)

ggplot(weights_long_df, aes(x = Asset, y = Weight, fill = Asset)) +
  geom_col() +
  facet_wrap(~ Portfolio) +
  scale_fill_manual(values = qvrs_colors) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Optimal Portfolio Allocation by Asset",
    subtitle = "Comparison between Global Minimum Variance and Maximum
    ↪ Sharpe portfolios",
    x = NULL, y = "Portfolio Weight"
  ) +
  theme_quant() +
  theme(legend.position = "none")

```

Optimal Portfolio Allocation by Asset

Comparison between Global Minimum Variance and Maximum Sharpe portfolios

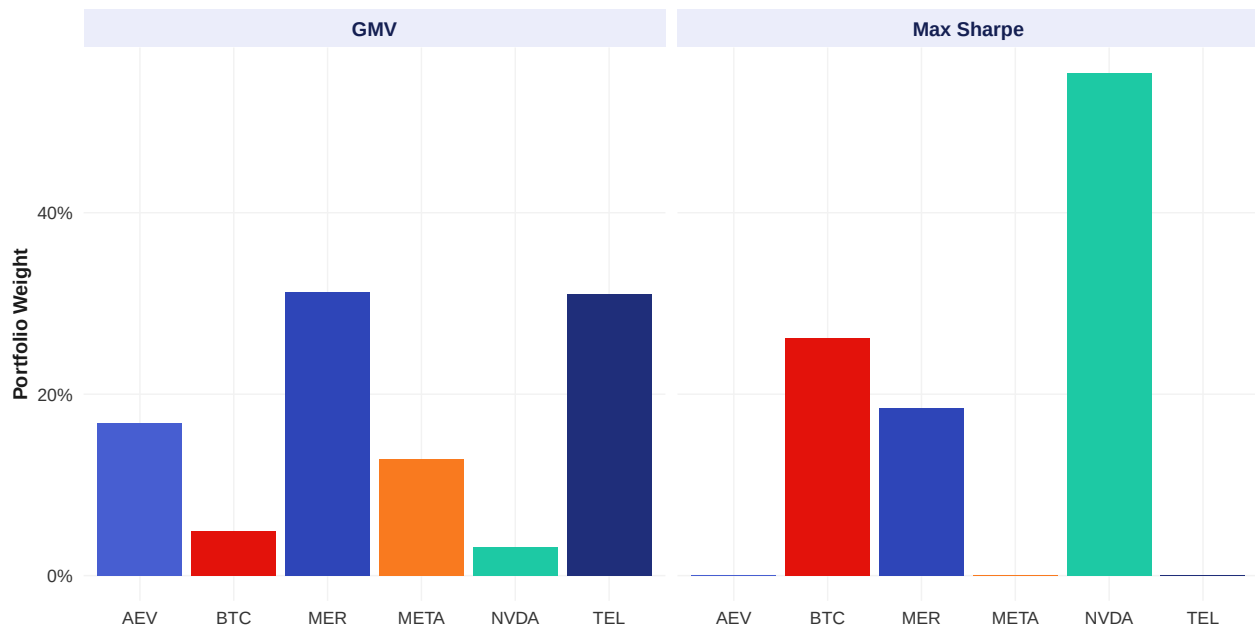


Figure 11: Optimal Portfolio Allocation by Asset (GMV vs Max Sharpe)

```

# Running-peak drawdown dates: depth measured from the running maximum
↪ wealth
# preceding the deepest trough, matching the reported max drawdown
↪ magnitudes.
dd_dates <- function(r_series) {

```

```

w <- cumprod(1 + as.numeric(r_series))
runmax <- cummax(w)
dd <- w / runmax - 1
t <- which.min(dd)
p <- which.max(w[1:t])
c(depth = sprintf("%.2f%%", dd[t] * 100), peak =
  ↪ format(index(r_series)[p]), trough = format(index(r_series)[t]))
}

bnh_dd <- dd_dates(r_bnh)
reb_dd <- dd_dates(r_reb)

nvda_peak_w <- max(weights_bnh[, "NVDA"])
nvda_peak_d <- format(index(weights_bnh)[which.max(weights_bnh[, "NVDA"])])

ann_vol_sigma <- sqrt(diag(Sigma_ann))

cat("Buy & Hold max drawdown:", bnh_dd["depth"], "peak", bnh_dd["peak"],
  ↪ "trough", bnh_dd["trough"], "\n")

```

Buy & Hold max drawdown: -58.21% peak 2021-11-08 trough 2022-10-14

```

cat("Monthly Rebalanced max drawdown:", reb_dd["depth"], "peak",
  ↪ reb_dd["peak"], "trough", reb_dd["trough"], "\n")

```

Monthly Rebalanced max drawdown: -43.20% peak 2021-12-27 trough 2022-10-12

```

cat("Buy & Hold NVDA weight peak:", sprintf("%.2f%%", nvda_peak_w * 100),
  ↪ "on", nvda_peak_d, "\n")

```

Buy & Hold NVDA weight peak: 62.29% on 2025-12-26

```

cat("Annualized volatility from optimization covariance (simple daily
  ↪ returns x 252):", paste(sprintf("%s %.2f%%", names(ann_vol_sigma),
  ↪ ann_vol_sigma * 100), collapse = ", "), "\n")

```

Annualized volatility from optimization covariance (simple daily returns x
 ↪ 252): TEL 31.13%, MER 31.38%, AEV 38.91%, NVDA 53.18%, META 43.82%, BTC
 ↪ 61.39%

8 Portfolio Risk Assessment (VaR, CVaR, Monte Carlo, Stress Testing)

```

port_returns_gmv <- Return.portfolio(

```

```

R = asset_returns_smp,
weights = w_gmv,
rebalance_on = "months"
)

```

```

port_returns_sharpe <- Return.portfolio(
  R = asset_returns_smp,
  weights = w_sharpe,
  rebalance_on = "months"
)

```

```

v_hist_gmv_95 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.95, method =
  ↪ "historical"))))
v_hist_gmv_99 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.99, method =
  ↪ "historical"))))
v_para_gmv_95 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.95, method =
  ↪ "gaussian"))))
v_para_gmv_99 <- abs(as.numeric(VaR(port_returns_gmv, p = 0.99, method =
  ↪ "gaussian"))))

```

```

e_hist_gmv_95 <- abs(as.numeric(ES(port_returns_gmv, p = 0.95, method =
  ↪ "historical"))))
e_hist_gmv_99 <- abs(as.numeric(ES(port_returns_gmv, p = 0.99, method =
  ↪ "historical"))))
e_para_gmv_95 <- abs(as.numeric(ES(port_returns_gmv, p = 0.95, method =
  ↪ "gaussian"))))
e_para_gmv_99 <- abs(as.numeric(ES(port_returns_gmv, p = 0.99, method =
  ↪ "gaussian"))))

```

```

v_hist_shp_95 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.95, method =
  ↪ "historical"))))
v_hist_shp_99 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.99, method =
  ↪ "historical"))))
v_para_shp_95 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.95, method =
  ↪ "gaussian"))))
v_para_shp_99 <- abs(as.numeric(VaR(port_returns_sharpe, p = 0.99, method =
  ↪ "gaussian"))))

```

```

e_hist_shp_95 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.95, method =
  ↪ "historical"))))
e_hist_shp_99 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.99, method =
  ↪ "historical"))))
e_para_shp_95 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.95, method =
  ↪ "gaussian"))))

```

```

e_para_shp_99 <- abs(as.numeric(ES(port_returns_sharpe, p = 0.99, method =
  ↪ "gaussian"))))

risk_summary_df <- data.frame(
  Portfolio = rep(c("GMV", "Max Sharpe"), each = 4),
  Metric    = rep(c("Historical VaR", "Parametric VaR", "Historical ES
  ↪ (CVaR)", "Parametric ES (CVaR)"), 2),
  `95% CI`  = sprintf("%.2f%%", c(
    v_hist_gmv_95, v_para_gmv_95, e_hist_gmv_95, e_para_gmv_95,
    v_hist_shp_95, v_para_shp_95, e_hist_shp_95, e_para_shp_95
  ) * 100),
  `99% CI`  = sprintf("%.2f%%", c(
    v_hist_gmv_99, v_para_gmv_99, e_hist_gmv_99, e_para_gmv_99,
    v_hist_shp_99, v_para_shp_99, e_hist_shp_99, e_para_shp_99
  ) * 100),
  check.names = FALSE
)

safe_kable(risk_summary_df, caption = "Historical and Parametric VaR / CVaR
  ↪ Summary (Daily, 95% and 99% Confidence)")

```

Table 8: Historical and Parametric VaR / CVaR Summary (Daily, 95% and 99% Confidence)

Portfolio	Metric	95% CI	99% CI
GMV	Historical VaR	1.80%	3.27%
GMV	Parametric VaR	2.03%	2.90%
GMV	Historical ES (CVaR)	2.85%	4.67%
GMV	Parametric ES (CVaR)	2.56%	3.33%
Max Sharpe	Historical VaR	3.39%	5.64%
Max Sharpe	Parametric VaR	3.74%	5.38%
Max Sharpe	Historical ES (CVaR)	5.11%	8.55%
Max Sharpe	Parametric ES (CVaR)	4.74%	6.19%

```

# Simulate 10,000 one-day return scenarios using multivariate normal
  ↪ distribution
set.seed(123)
n_sims <- 10000

mu_daily    <- mu_ann / 252
Sigma_daily <- Sigma_ann / 252

sim_asset_returns <- mvrnorm(n = n_sims, mu = mu_daily, Sigma = Sigma_daily)
colnames(sim_asset_returns) <- asset_names

sim_port_returns_gmv <- as.numeric(sim_asset_returns %*% w_gmv)

```

```

sim_port_returns_sharpe <- as.numeric(sim_asset_returns %*% w_sharpe)

mc_var_gmv_95      <- quantile(sim_port_returns_gmv, 0.05)
mc_var_sharpe_95 <- quantile(sim_port_returns_sharpe, 0.05)

mc_es_gmv_95      <- mean(sim_port_returns_gmv[sim_port_returns_gmv <=
  ↪ mc_var_gmv_95])
mc_es_sharpe_95 <- mean(sim_port_returns_sharpe[sim_port_returns_sharpe <=
  ↪ mc_var_sharpe_95])

mc_summary_df <- data.frame(
  Portfolio = c("GMV", "Max Sharpe"),
  `MC VaR (95%)` = sprintf("%.2f%%", c(abs(mc_var_gmv_95),
  ↪ abs(mc_var_sharpe_95)) * 100),
  `MC ES (95%)` = sprintf("%.2f%%", c(abs(mc_es_gmv_95),
  ↪ abs(mc_es_sharpe_95)) * 100),
  check.names = FALSE
)

```

```

safe_kable(mc_summary_df, caption = "Monte Carlo Simulated VaR and Expected
  ↪ Shortfall (10,000 Simulations)")

```

Table 9: Monte Carlo Simulated VaR and Expected Shortfall (10,000 Simulations)

Portfolio	MC VaR (95%)	MC ES (95%)
GMV	2.05%	2.55%
Max Sharpe	3.66%	4.71%

```

dist_compare_df <- bind_rows(
  data.frame(Return = as.numeric(port_returns_sharpe), Source =
  ↪ "Historical"),
  data.frame(Return = sim_port_returns_sharpe, Source = "Monte Carlo")
)

ggplot(dist_compare_df, aes(x = Return, fill = Source, color = Source)) +
  geom_density(alpha = 0.35, linewidth = 0.7) +
  scale_fill_manual(values = c("Historical" = "#1F2E7A", "Monte Carlo" =
  ↪ "#E3120B")) +
  scale_color_manual(values = c("Historical" = "#1F2E7A", "Monte Carlo" =
  ↪ "#E3120B")) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  labs(
    title = "Historical vs Monte Carlo Simulated Return Distribution",
    subtitle = "Max Sharpe Portfolio: comparing observed daily returns to
    ↪ 10,000 simulated draws",

```

```

x = "Daily Portfolio Return",
y = "Density"
) +
theme_quant()

```

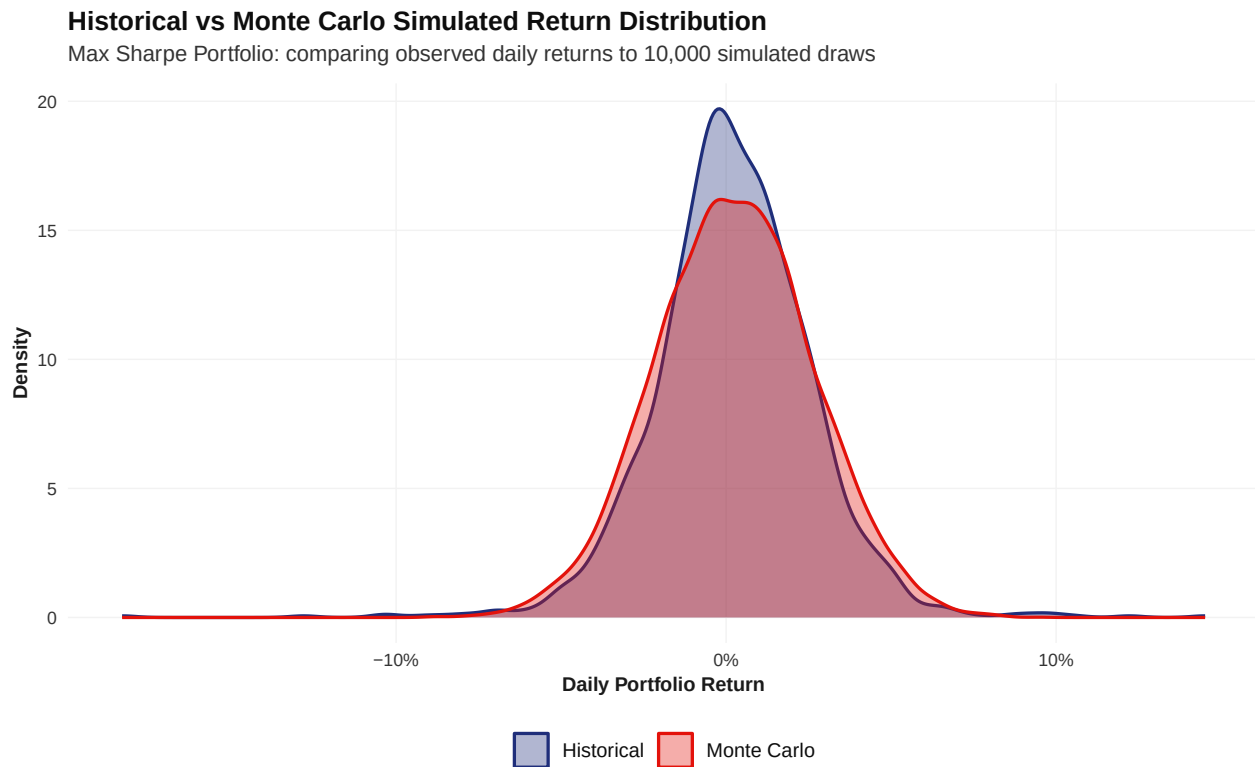


Figure 12: Historical vs Monte Carlo Simulated Return Distribution (Max Sharpe Portfolio)

```

# Historical Stress Scenario Windows
covid_crash_window    <- "2020-02-19/2020-03-23"
tech_selloff_window   <- "2021-11-19/2022-01-27"
crypto_winter_window  <- "2022-05-05/2022-05-13"
fed_hike_window       <- "2022-06-08/2022-06-16"

calc_window_shock <- function(ret_xts, date_range) {
  window_returns <- ret_xts[date_range]
  sapply(window_returns, function(col) prod(1 + col, na.rm = TRUE) - 1)
}

shock_covid    <- calc_window_shock(ret_smp, covid_crash_window)
shock_tech     <- calc_window_shock(ret_smp, tech_selloff_window)
shock_crypto   <- calc_window_shock(ret_smp, crypto_winter_window)
shock_fed      <- calc_window_shock(ret_smp, fed_hike_window)

```

```

worst_observed <- pmin(shock_covid, shock_crypto, shock_fed, shock_tech)
shock_black_swan <- worst_observed * 1.5

stress_scenarios <- tibble(
  Scenario = c("2020 COVID Crash", "2022 Crypto Winter", "Fed Rate Hike
    ↳ Shock",
               "Tech Sector Selloff", "Hypothetical Black Swan"),
  TEL = c(shock_covid["TEL"], shock_crypto["TEL"], shock_fed["TEL"],
    ↳ shock_tech["TEL"], shock_black_swan["TEL"]),
  MER = c(shock_covid["MER"], shock_crypto["MER"], shock_fed["MER"],
    ↳ shock_tech["MER"], shock_black_swan["MER"]),
  AEV = c(shock_covid["AEV"], shock_crypto["AEV"], shock_fed["AEV"],
    ↳ shock_tech["AEV"], shock_black_swan["AEV"]),
  NVDA = c(shock_covid["NVDA"], shock_crypto["NVDA"], shock_fed["NVDA"],
    ↳ shock_tech["NVDA"], shock_black_swan["NVDA"]),
  META = c(shock_covid["META"], shock_crypto["META"], shock_fed["META"],
    ↳ shock_tech["META"], shock_black_swan["META"]),
  BTC = c(shock_covid["BTC"], shock_crypto["BTC"], shock_fed["BTC"],
    ↳ shock_tech["BTC"], shock_black_swan["BTC"])
)

```

```

stress_impact_df <- stress_scenarios %>%
  rowwise() %>%
  mutate(
    shock_vec      = list(c(TEL, MER, AEV, NVDA, META, BTC)),
    GMV_Impact     = sum(unlist(shock_vec) *
    ↳ w_gmv[c("TEL", "MER", "AEV", "NVDA", "META", "BTC")]),
    MaxSharpe_Impact = sum(unlist(shock_vec) *
    ↳ w_sharpe[c("TEL", "MER", "AEV", "NVDA", "META", "BTC")])
  ) %>%
  ungroup() %>%
  dplyr::select(Scenario, GMV_Impact, MaxSharpe_Impact)

```

```

stress_summary_df <- stress_impact_df %>%
  mutate(
    GMV_Impact = sprintf("%.2f%%", GMV_Impact * 100),
    MaxSharpe_Impact = sprintf("%.2f%%", MaxSharpe_Impact * 100)
  ) %>%
  rename(`GMV Portfolio Impact` = GMV_Impact, `Max Sharpe Portfolio Impact`
    ↳ = MaxSharpe_Impact)

safe_kable(stress_summary_df, caption = "Portfolio Impact Under Stress
  ↳ Scenarios")

```

Table 10: Portfolio Impact Under Stress Scenarios

Scenario	GMV Portfolio Impact	Max Sharpe Portfolio Impact
2020 COVID Crash	-22.55%	-28.83%
2022 Crypto Winter	-4.90%	-14.40%
Fed Rate Hike Shock	-9.47%	-18.65%
Tech Sector Selloff	3.71%	-24.58%
Hypothetical Black Swan	-35.89%	-45.30%

```

stress_long_df <- stress_impact_df %>%
  pivot_longer(cols = c(GMV_Impact, MaxSharpe_Impact), names_to =
    ↪ "Portfolio", values_to = "Impact") %>%
  mutate(Portfolio = ifelse(Portfolio == "GMV_Impact", "GMV", "Max Sharpe"))

ggplot(stress_long_df, aes(x = reorder(Scenario, Impact), y = Impact, fill
  ↪ = Portfolio)) +
  geom_col(position = "dodge") +
  coord_flip() +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  scale_fill_manual(values = c("GMV" = "#1F2E7A", "Max Sharpe" = "#E3120B"))
  ↪ +
  labs(
    title = "Portfolio Impact Under Stress Scenarios",
    subtitle = "Simulated instantaneous shock to portfolio value, GMV vs Max
    ↪ Sharpe weighting",
    x = NULL,
    y = "Portfolio Return Impact"
  ) +
  theme_quant()

```

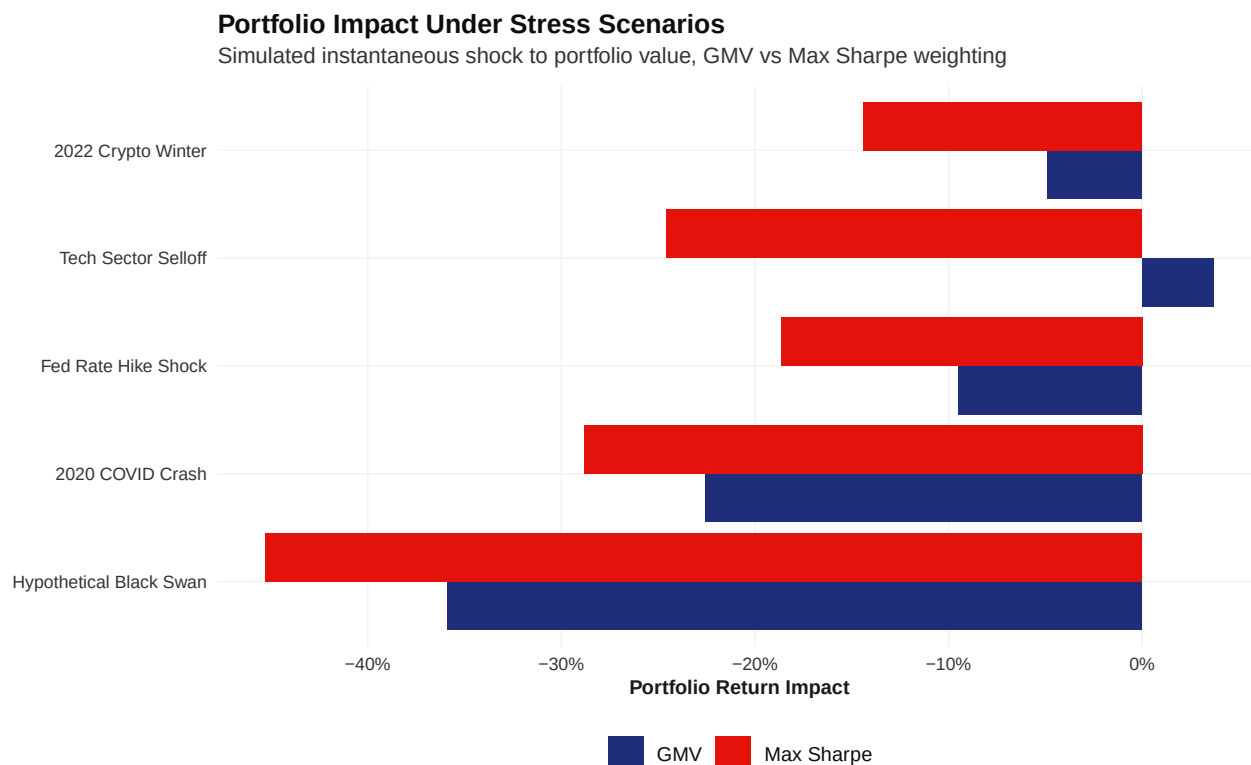


Figure 13: Portfolio Value Impact Across Stress Scenarios

9 Executive Investment Recommendation

Based on the portfolio optimization, historical performance, downside-risk measures, and stress-testing results, the Maximum Sharpe Ratio portfolio is recommended. The portfolio allocates 55.44 percent to NVIDIA (NVDA), 26.14 percent to Bitcoin (BTC), and 18.42 percent to Manila Electric Company (MER), delivering an expected annual return of 58.10 percent, an annualized volatility of 38.25 percent, and a Sharpe ratio of 1.3815 at the 5.25 percent risk-free rate, the highest risk-adjusted profile in the feasible universe. Figure 11 contrasts this allocation with the GMV portfolio, and Figure 10 locates it at the tangency point of the efficient frontier.

NVIDIA contributes the strongest sample-period expected return (71.65 percent annualized) and, together with Bitcoin (61.49 percent annualized), drives the portfolio's return profile. Bitcoin's inclusion rests on its high expected return and its low correlation with the Philippine constituents (MER–BTC 0.0286), while NVIDIA pairs with the domestic block at near-zero correlation (MER–NVDA 0.0121, MER–META 0.0172) and at moderate co-movement with Bitcoin (NVDA–BTC 0.3204). Manila Electric Company anchors the allocation as the sole domestic constituent, combining a positive expected return (12.49 percent annualized) with near-zero correlation to both growth assets. TEL, AEV, and META receive zero weight in the long-only tangency solution.

Downside risk is governed by the empirical tail measures reported in Table 8. The recommended portfolio's daily 95 percent Historical VaR is -3.39 percent and its 95 percent Historical Expected Shortfall is -5.11 percent; at the 99 percent level these reach -5.64 percent and -8.55 percent. The empirical 99 percent Expected Shortfall (8.55 percent) exceeds the Gaussian parametric estimate (6.19 percent) by 236 basis points per day, so historical measures, rather than the normal approximation, should set the risk budget.

Stress testing confirms the trade-off between return and capital preservation. Under the five scenarios in Figure 13 and Table 10, the Maximum Sharpe portfolio loses 28.83 percent in the 2020 COVID crash, 24.58 percent in the technology-sector selloff, 18.65 percent in the 2022 rate-hike shock, and 14.40 percent in the 2022 crypto winter, with a -45.30 percent loss under the hypothetical 1.5x Black Swan scenario. The GMV portfolio contains the same shocks at materially smaller magnitudes (-22.55 percent COVID, -35.89 percent Black Swan), illustrating its role as the defensive alternative.

Disciplined monthly rebalancing is required to manage the portfolio's concentration: NVDA and BTC together represent 81.58 percent of the recommended allocation, and un-rebalanced equal-weight exposure drifted to 61.75 percent NVDA and 24.77 percent BTC by December 2025. Over the sample, monthly rebalancing reduced volatility from 35.12 percent to 24.18 percent and the peak drawdown from -58.21 percent to -43.20 percent at the cost of cumulative return (384.85 percent versus 743.04 percent). For investors prioritizing capital preservation, the GMV portfolio allocates TEL 31.01 percent, MER 31.23 percent, AEV 16.85 percent, NVDA 3.15 percent, META 12.86 percent, and BTC 4.90 percent, delivering a 16.03 percent expected return at 20.22 percent volatility (Sharpe 0.5332). Portfolio choice ultimately reflects the investor's willingness and capacity to bear losses: the Maximum Sharpe portfolio suits investors seeking higher returns despite greater risk, while the GMV portfolio suits investors who cannot tolerate peak stress losses near -45 percent.

10 References

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- Fabozzi, F. J., Focardi, S. M., & Kolm, P. N. (2002). *Financial Modeling of the Equity Market: From Portfolio Optimization to Stress Testing*. John Wiley & Sons.
- Jorion, P. (2007). *Value at Risk: The New Benchmark for Managing Financial Risk* (3rd ed.). McGraw-Hill.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77-91.
- Sharpe, W. F. (1964). Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3), 425-442.

11 Appendix: Complete R Analysis Code

```
# FINSTAD Case 2 Group 3 (Charlie) Complete Portfolio & Risk Pipeline

library(tidyverse)
library(lubridate)
library(scales)
library(xts)
library(zoo)
library(PerformanceAnalytics)
library(nloptr)
library(quadprog)
library(MASS)

raw_data <- read_csv("../data/master_dataset.csv", show_col_types = FALSE)
→ %>%
  mutate(Date = as.Date(Date)) %>%
  filter(Date >= as.Date("2020-01-02") & Date <= as.Date("2025-12-31")) %>%
  arrange(Date)

asset_cols <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC", "SPY")
portfolio_assets <- c("TEL", "MER", "AEV", "NVDA", "META", "BTC")

prices_mat <- as.matrix(raw_data[, asset_cols])
prices_xts <- xts(prices_mat, order.by = raw_data[["Date"]])
prices_carried <- na.locf(prices_xts)

returns_xts_carried <- Return.calculate(prices_carried, method = "log")[-1,
→ ]
asset_returns_carried <- returns_xts_carried[, portfolio_assets]

ret_smp <- exp(returns_xts_carried) - 1
asset_returns_smp <- ret_smp[, portfolio_assets]

w_eq <- rep(1 / 6, 6)
names(w_eq) <- portfolio_assets

port_bnh <- Return.portfolio(R = asset_returns_smp, weights = w_eq)
port_reb <- Return.portfolio(R = asset_returns_smp, weights = w_eq,
→ rebalance_on = "months")

mu_ann <- colMeans(asset_returns_smp, na.rm = TRUE) * 252
Sigma_ann <- cov(asset_returns_smp, use = "pairwise.complete.obs") * 252
rf_annual <- 0.0525
```

```

eval_f_var <- function(w) as.numeric(t(w) %*% Sigma_ann %*% w)
eval_grad_f_var <- function(w) nl.grad(w, eval_f_var)
eval_g_eq <- function(w) sum(w) - 1
eval_jac_g_eq <- function(w) nl.jacobian(w, eval_g_eq)

w0 <- rep(1/6, 6)
gmw_sol <- nloptr(x0 = w0, eval_f = eval_f_var, eval_grad_f =
  ↪ eval_grad_f_var,
               lb = rep(0,6), ub = rep(1,6), eval_g_eq = eval_g_eq,
               eval_jac_g_eq = eval_jac_g_eq,
               opts = list("algorithm" = "NLOPT_LD_SLSQP", "xtol_rel" =
  ↪ 1e-8))
w_gmw <- gmw_sol$solution
names(w_gmw) <- portfolio_assets

eval_f_sharpe <- function(w) {
  p_ret <- sum(w * mu_ann)
  p_sd <- sqrt(as.numeric(t(w) %*% Sigma_ann %*% w))
  -(p_ret - rf_annual) / p_sd
}
eval_grad_f_sharpe <- function(w) nl.grad(w, eval_f_sharpe)

shp_sol <- nloptr(x0 = w0, eval_f = eval_f_sharpe, eval_grad_f =
  ↪ eval_grad_f_sharpe,
               lb = rep(0,6), ub = rep(1,6), eval_g_eq = eval_g_eq,
               eval_jac_g_eq = eval_jac_g_eq,
               opts = list("algorithm" = "NLOPT_LD_SLSQP", "xtol_rel" =
  ↪ 1e-8))
w_sharpe <- shp_sol$solution
names(w_sharpe) <- portfolio_assets

port_ret_gmw <- Return.portfolio(R = asset_returns_smp, weights = w_gmw,
  ↪ rebalance_on = "months")
port_ret_shp <- Return.portfolio(R = asset_returns_smp, weights = w_sharpe,
  ↪ rebalance_on = "months")

var_gmw_95 <- abs(VaR(port_ret_gmw, p = 0.95, method = "historical"))
var_shp_95 <- abs(VaR(port_ret_shp, p = 0.95, method = "historical"))
es_gmw_95 <- abs(ES(port_ret_gmw, p = 0.95, method = "historical"))
es_shp_95 <- abs(ES(port_ret_shp, p = 0.95, method = "historical"))

```